

Technical Validation Supplement

Measurement of $\nu_\mu + \bar{\nu}_\mu$ Charged-Current Single Pion Production on Argon in the NuMI Beam

Companion to the analysis note

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Abstract

This supplement holds the validation and diagnostic material for the $\text{CC}\pi^\pm$ NuMI cross-section measurement: the studies that establish what the measurement can and cannot resolve, the estimators that were tried and rejected, and the closure tests that qualify the extraction. It is separated from the analysis note so that the note can state the analysis and its results while the studies that justify its choices remain available and citable at full length.

The two documents are checked for numerical consistency where current results are duplicated (`report/check_tables.py`, `xsec_analyzer/check_staleness.py`); every number here is from the release named on the title page unless a table says otherwise. The history of the analysis, the corrections that shaped it and the values before and after, is in the change log and is not repeated here.

The central results of this supplement are negative ones, and they set the scope of the measurement. Charged pions interact hadronically before ranging out, so the reconstructed momentum saturates near 0.34 GeV/ c and carries little information about the initial momentum above roughly 0.4 GeV/ c . Every alternative estimator tried — golden-pion tagging, multiple Coulomb scattering, calorimetry, multivariate regression, daughter-energy recovery, and the proton-tagged selection — fails to recover it, and a truth-level golden selection performs no better than the classifier, which places the limitation in the physics rather than in the estimator. This is what reduces p_π to two bins and $W_{\pi p}$ to an integrated cross section.

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1 Pion momentum: the range estimator and why it fails

1.1 The pion-momentum bias and why p_π is the worst-conditioned observable

The p_π response is the least diagonal of any observable in the analysis. This section quantifies why, because the cause is physical and sets a floor on what the measurement can say about the pion momentum spectrum.

Two separate effects, only one of them corrected. The reconstructed pion momentum stored in the ntuple, `candidate_pion_mom_reco`, is `trk_range_muon_mom_v`: a range-based momentum evaluated under the *muon* mass hypothesis and applied to a pion track. Two distinct things follow.

First, the mass hypothesis itself biases the estimate low, because for a given range a pion carries more momentum than a muon. The size is $\sim 17\text{--}19\%$ from the CSDA range scaling, and it is essentially independent of momentum. This effect *is* corrected, though not in the branch: the reco bin definitions in `ccpi_ppi_bin_config_opt.txt` apply a kinetic-energy-preserving $\mu \rightarrow \pi$ mass swap, $E_\mu \rightarrow T \rightarrow E_\pi \rightarrow p_\pi$, before binning. It works; in the lowest true bin $\langle p^{\text{reco}}/p^{\text{true}} \rangle$ goes from 0.888 with the raw branch to 0.976 with the correction.

Second, and not corrected by anything, charged pions interact hadronically before coming to rest. The reconstructed track then ends at the interaction point rather than the stopping point, so the visible range, and any momentum inferred from it, is truncated. This grows rapidly with momentum and comes to dominate.

The reconstructed momentum saturates. Table 1 and Figure 1 make the consequence plain: as the true momentum rises by a factor of four and a half, the reconstructed momentum barely moves, saturating at ≈ 0.34 GeV/ c . Above that the estimator carries almost no information about how energetic the pion actually was.

Table 1: Pion momentum reconstruction versus truth, selected signal, Run-1 FHC. “raw” is the `candidate_pion_mom_reco` branch; “corr.” additionally applies the $\mu \rightarrow \pi$ mass swap used by the reco bin definitions. The mass correction removes the bias at threshold; nothing removes the saturation above it.

true p_π [GeV/ c]	N	$\langle p^{\text{true}} \rangle$	$\langle p^{\text{corr}} \rangle$	raw/true	corr./true
0.175–0.250	528	0.210	0.205	0.888	0.976
0.250–0.300	270	0.276	0.238	0.790	0.863
0.300–0.350	254	0.325	0.253	0.716	0.779
0.350–0.400	265	0.374	0.270	0.664	0.721
0.400–0.500	403	0.443	0.279	0.581	0.629
0.500–0.600	252	0.548	0.307	0.520	0.560
0.600–0.800	272	0.688	0.326	0.441	0.473
0.800–1.200	164	0.951	0.336	0.329	0.353

What that does to the response matrix. Migrating the five analysis bins with the corrected estimator gives a nearly lower-triangular matrix: the diagonal fraction falls from 94.3% in the first true bin to 36.5, 23.1, 10.4 and finally 6.2% in the last. For the highest true bin ($p_\pi > 0.55$ GeV/ c) only 6.2% of events are reconstructed in the corresponding reco bin while 32.6% fall all the way into the *lowest* one. Every true bin above the first therefore deposits most of its events into reco bin 1, which is exactly the “pile-up into the first bin” seen in Fig. 17 of the analysis note.



Figure 1: Mean reconstructed against mean true pion momentum, selected signal. The analysis estimator (filled, $\mu \rightarrow \pi$ mass corrected) tracks truth at threshold but flattens out near 0.34 GeV/c: a true 0.95 GeV/c pion reconstructs at the same place as a true 0.55 GeV/c one. The raw `candidate_pion_mom_reco` branch (open) sits systematically below it by the uncorrected mass-hypothesis offset. The saturation is caused by hadronic interaction truncating the track range, and no range-based estimator can recover it.

How the BNB $\text{CC}1\pi^\pm$ analysis handles the same problem. The published BNB $\text{CC}1\pi^\pm\text{Xp}$ analysis (internal note v0.91) faces an identical estimator, it too takes “the momentum of pion candidates from range”, and its treatment is worth reproducing here, because it is the recipe this analysis followed.

That analysis does three things we do not. First, it defines a *golden pion*: one that comes to rest and leaves a Bragg peak, as opposed to one that scattered or interacted. Second, it trains a dedicated *golden-pion BDT* and applies it as an extra cut for the pion-momentum differential cross section only, leaving the total and angular results on the generic selection. The BDT separates golden from non-golden pions using track “wiggleness” (the standard deviation of the angular difference between directions of adjacent points along the fitted track, so a direct handle on scattering), the number of reconstructed descendant particles, the track-shower score, and the Bragg likelihood ratio. The cost and benefit are quoted explicitly: efficiency falls from 20.3% to 9.5% and purity is essentially unchanged (51.9 $\rightarrow$ 49.5%), but the golden fraction of the selected signal rises from 36.3% to 67.7%. They apply the same logic to the muon momentum, restricting that cross section to contained muons (8.4% efficiency, 56.1% purity) so it uses range only.

Third, and most directly comparable, their binning is chosen against an explicit criterion: each bin must hold at least 100 predicted MC events *and* the diagonal element of the smearing matrix must exceed 0.68. The resulting pion-momentum edges are 0.10, 0.16, 0.19, 0.22, 0.60 GeV/c with an overflow bin, fine structure only where the estimator still tracks truth, then a single wide bin from 0.22 to 0.60 and everything above 0.60 in overflow.

Set against that criterion our binning does not survive: the diagonal fractions measured above are 94.3, 36.5, 23.1, 10.4 and 6.2%, so *one* of five bins passes 0.68 and the top three fail it by a wide margin. Our edges (0.175–0.550 in five bins, open above) put four boundaries inside precisely the region where the reconstructed momentum has saturated.

Consequences, and what would improve it. This is not a bias on the extracted cross section; the response matrix contains it and the unfolding inverts it, but it is the reason the p_π result leans harder on the regularisation than any other observable. With 6% diagonal strength in the top bin, the measurement above ~ 0.5 GeV/ c is effectively inferred from the response model rather than measured.

The BNB note therefore supplies a tested recipe, and *all four* of its BDT inputs are already available to us. The Bragg likelihood ratios (`trk_bragg_pion`, `_mip`, `_mu`, `_p`) and the track-shower score are in the ntuples; the multi-pion pion-ID BDT in this analysis already trains on exactly those, and the descendant counts are `pfp_trk_daughters_v` / `pfp_shr_daughters_v`. Wiggleness is there too, under a different name: the PeLEE ntuples carry `trk_avg_deflection_stdev_v` (with companions `_mean_v` and `_separation_mean_v`), which is precisely the standard deviation of the angular deflection between adjacent track points. It is fully populated and is carried forward by `ProcessNTuples`.

Wiggleness works, but with the opposite sign to the naive expectation. Testing it directly on truth-matched charged pions in the Run-1 FHC overlay, calling a pion “golden-like” when the mass-corrected range momentum recovers the true momentum to within 20%, and “interacting” when it recovers less than 60%, gives a clear separation, but golden pions are the *wigglier* ones, by about a factor two in the mean (0.0048 against 0.0027). That is not a momentum artefact: it survives binning in track length, with the ratio staying at 0.51–0.63 from 5 cm to 400 cm.

The sign makes physical sense on reflection. A pion that stops travels its full range and scatters violently through its final slow centimetres, so the deflection spread averaged along the track is large. A pion that interacts hadronically is cut short while still fast and never reaches that wiggly end, so its track is straighter. High wiggleness is thus a signature of *having stopped*, exactly the population for which a range-based momentum is meaningful.

As a single cut it is already useful: on this sample the golden fraction is 39.9% before any cut, and requiring `trk_avg_deflection_stdev_v` > 0.001 keeps 86.8% of golden pions while rejecting 48.8% of interacting ones, lifting the golden fraction to 53.0%. A tighter cut at 0.002 gives 57.8% efficiency and 55.7% golden. For reference the BNB’s four-variable BDT reaches 67.7% golden at roughly half the generic-selection efficiency, so one variable recovers a good part of the gain and the remaining three, all of which we have, should close much of the rest.

Three options follow from this, in increasing order of effort: rebin p_π against the BNB’s own criterion (diagonal > 0.68), which on the numbers above means one resolved bin at threshold plus a wide open bin; add a golden-pion cut for the p_π cross section built from the variables already available, accepting the factor-two efficiency loss the BNB analysis accepts; or reproduce their BDT in full. None changes the total or angular results, which do not use this estimator. The first is the adopted scheme (§1.1); the classifier was trained and tested and found not to change the conclusion (§1.1).

Adopted response: a two-bin p_π scheme

The released binning is two bins, [0.175, 0.205] GeV/ c and everything above, which raises the migration diagonals from 96/35/21/10/6% (full analysis sample; 94.3/36.5/23.1/10.4/6.2% on Run 1 alone, Table 1) to **91/76%**, clearing the 68% criterion in *both* bins. (These are measured on the analysis sample; the golden-pion study of Sec. 1.1 uses a looser track sample whose diagonals are lower, and the two sets are not interchangeable.)

The boundary is not a free choice dressed up as an optimisation. The worst diagonal is always the upper bin and it falls monotonically as the boundary rises, so the binding constraint is the *width* of the first bin rather than the diagonal itself:

boundary	first-bin width	diagonal, bin 1	diagonal, bin 2
0.185	10 MeV/ c	79.7%	85.4%
0.195	20 MeV/ c	88.8%	80.0%
0.200	25 MeV/ c	90.4%	77.7%
0.205	30 MeV/c	91.0%	75.5%
0.220	45 MeV/ c	93.3%	69.1%
0.250	75 MeV/ c	95.6%	56.1%
0.300	125 MeV/ c	96.3%	39.0%

Both bins clear the criterion anywhere between roughly 0.19 and 0.22, so the choice within that window is governed by bin width rather than by the criterion. 0.205 is the narrowest first bin with precedent: 30 MeV/ c matches the narrowest bin used in the BNB CC1 π analysis. Going narrower trades first-bin diagonal for upper-bin diagonal and at 10 MeV/ c the first bin itself falls to 80% on only 325 events, which is not a useful cross-section bin; going wider, 0.220 still passes but leaves the upper bin at 69%, with no margin. An exhaustive scan over all edge sets (`macros/ppi_binning_scan.C`) returns 0.205 as the best first boundary for every number of bins tried, and for the proton-tagged selection as well.

A three-bin variant was built and measured before being discarded. The best achievable (same 30 MeV/ c minimum width, edges 0.205 and 0.265) is 91/51/52%, against 91/76% for two bins: the third bin costs 25 points on the worst bin and drops it well below the criterion. Two 30 MeV/ c bins followed by an open bin (0.205, 0.235) is worse still at 91/39/62%, and note that it degrades the *upper* bin too, from 76% to 62%: narrowing the top bin's reco range pushes genuinely high-momentum events out of their own bin. Four bins reach only 91/39/39/42%. Above ~ 0.25 GeV/ c the reconstructed momentum barely varies, so a second boundary there separates almost nothing and merely divides one poorly-measured region into two equally poor halves. The same ordering holds under a golden-pion cut (67/52/46%), so it is not an artefact of the selection.

Closure. The two-bin scheme is carried through the full chain, universe construction, Wiener-SVD unfolding and systematics, and closes in all three configurations (Table 2); a five-bin extraction on the same sample closes too, at $\chi^2 \approx 2/5$, which is the point: closure cannot expose a response that carries no shape information, only the migration criterion can.

Table 2: Two-bin p_π result for all three beam configurations, $d\sigma/dp_\pi$ in 10^{-38} cm²/(GeV/ c)/Ar. Bin 1 is [0.175, 0.205] GeV/ c , bin 2 everything above. All curves are A_C -smeared, so they live in the same space as the measurement. Generated from the 2026-09-06 release (`report/tools/ppi2bin_tables.py`: closure sidecars and `curves_incl*_ppi2bin.tsv`); the uncertainty row is the total.

	FHC			RHC			Combined		
	bin 1	bin 2	σ_{int}	bin 1	bin 2	σ_{int}	bin 1	bin 2	σ_{int}
unfolded data	1.96	0.96	0.822	1.79	0.87	0.748	2.06	0.90	0.775
uncertainty	± 1.36	± 0.37		± 1.12	± 0.34		± 1.26	± 0.34	
A_C truth	2.34	0.95		1.71	0.89		2.17	0.90	
uB tune	2.29	0.93		1.72	0.88		2.15	0.89	
GENIE	1.82	0.75		1.29	0.63		1.67	0.67	
GiBUU	3.28	0.94		2.27	0.82		3.03	0.85	
NEUT	3.09	1.18		2.24	1.02		2.87	1.07	
NuWro	2.98	1.12		2.11	0.96		2.74	1.01	
χ^2/ndf	0.11/2 ($p = 0.95$)			0.01/2 ($p = 0.99$)			0.01/2 ($p = 1.00$)		

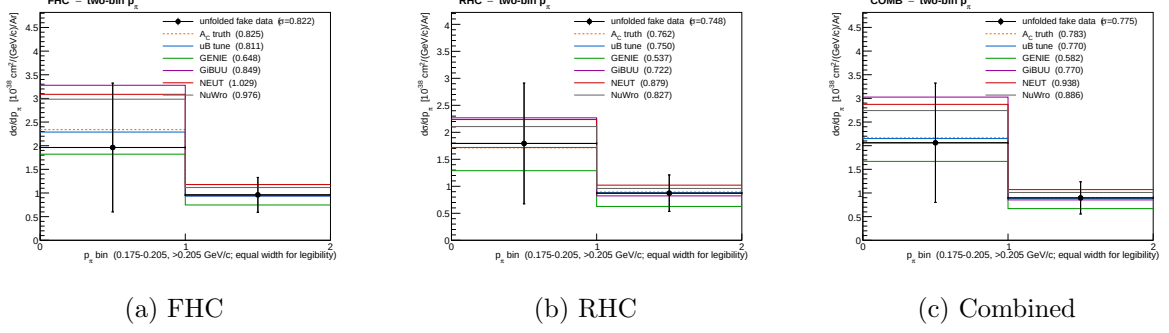


Figure 2: The two-bin p_π cross section, the graphical form of Tab. 2. Unfolded fake data (points, with the full systematic uncertainty) against the A_C -smeared truth, the uB tune and the four generators; every curve is A_C -smeared, so all of them live in the same space as the measurement. **The two bins are drawn with equal width for legibility:** they are 0.030 and 0.795 GeV/c wide, so to scale the first would occupy 4% of the axis. Area on these panels is therefore not proportional to cross section, and the integrated σ is printed on each panel instead. The generators bracket the data in every bin of every configuration; GENIE is generally the lowest prediction and NEUT and GiBUU the highest, and which generator is closest changes from bin to bin.

On the current release all three configurations close, with χ^2 of 0.11, 0.015 and 0.01 on two bins (the RHC value rounds either way in the note’s tables) and unfolded-to-truth integral ratios of 0.997, 0.981 and 0.989 (width-weighted unfolded over A_C -smeared truth from the released curves, `curves_incl*_ppi2bin.tsv`; the χ^2 from Tables 17–19 of the analysis note). The four generators bracket the data in both bins of every configuration, NEUT and GiBUU highest and GENIE lowest, the same ordering as in the other observables; the released A_C for FHC is $\begin{pmatrix} 0.53 & 0.01 \\ -0.51 & 0.87 \end{pmatrix}$ (`A_C_incl_FHC5_ppi2bin.tsv`), whose off-diagonal element is the reminder that the two bins are not independent after merging. The total fractional uncertainty is 50% (FHC), the largest of the inclusive observables, because the two bins each carry a large share of the flux normalisation with little statistical averaging. Figure 3 shows A_C for all three configurations (release 2026-09-06). RHC regularises a little harder than FHC in the first bin (diagonal 0.38 against 0.53, off-diagonal -0.37 against -0.51), and the combined extraction borrows the most (diagonal 0.53, off-diagonal -0.58); the second-bin diagonals are 0.87–0.88 in all three. Despite the different conditioning the three extractions close in normalisation within 2% (the ratios 0.997, 0.981 and 0.989 above), so there is no common normalisation offset associated with how hard A_C smears.

At 91/76% both bins clear the 68% criterion, so the two-bin scheme makes the inclusive p_π result usable on the standard test rather than merely honest about its limitations. It remains a coarse measurement: two bins cannot resolve a shape, and above ~ 0.34 GeV/c the reconstructed momentum saturates, so the upper bin is an integral over a wide range rather than a differential point. What the scheme buys is a result whose response is defensible, not a competitive shape measurement.

A golden-pion classifier was trained and tested: it works, and it is not enough

Following the BNB recipe, a golden-pion BDT was trained and evaluated (`macros/golden_pion_train.C`, `macros/golden_pion_compare.C`). The truth label is `mc_end_p < 0.01` GeV/c, the pion ranged out, joined to reconstruction by matching `backtracked_p` to `mc_p`, which pairs 100% of candidate tracks with no ambiguity. The inputs are the three deflection variables, the four Bragg likelihood ratios, the track–shower score and the descendant count. Training used 126 518 labelled tracks from Runs 1, 2 and 4; the split is by *run*, so all numbers below are on Run 2, which the classifier

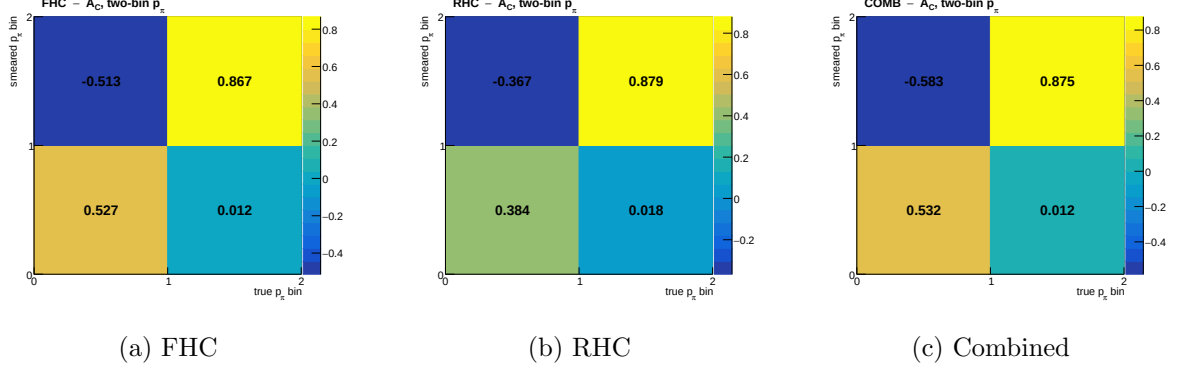


Figure 3: The additional-smearing matrix A_C for the two-bin p_π scheme, in the convention $\hat{x}_i = \sum_j (A_C)_{ij} t_j$ so that a column is a true bin and a row is the smeared bin it feeds. A_C is not normalisation-preserving, so its rows do not sum to unity. The negative entry (true bin 1 feeding smeared bin 2, drawn top-left since the smeared axis runs upward) is the regularisation borrowing between the two bins; it is why the truth curves in Fig. 2 must be A_C -smeared before being compared with the measurement rather than plotted raw.

never saw.

As a classifier it reproduces the BNB benchmark. ROC integral 0.758. At 44.8% efficiency it lifts the golden fraction from 43.1% to 64.9%, against the BNB’s $36.3 \rightarrow 67.7\%$ at roughly 47% efficiency. The variable ranking is dominated by the deflection variables (**wmean**, then **wstd**), with the Bragg ratios contributing essentially nothing, worth noting, since Bragg was expected to carry the stopping signature.

But it does not fix the binning, and cannot. Table 3 gives the migration diagonal in the five analysis bins as the cut is tightened. The first bin is already fine and stays fine; no other bin reaches 0.68 at any working point, even at 15% efficiency.

Table 3: Golden-pion BDT: efficiency, achieved golden fraction, and the p_π migration diagonal per true bin, on the held-out run. The BNB criterion for a usable bin is a diagonal above 0.68.

BDT cut	eff.	golden frac.	b1	b2	b3	b4	b5
none	100%	43.1%	99.2	23.7	13.2	6.4	5.5
> -0.05	57.0%	58.1%	99.1	29.4	19.7	11.2	7.1
> 0.00	44.8%	64.9%	99.1	33.1	24.5	15.4	9.1
$> +0.10$	24.6%	79.4%	99.1	42.9	36.8	29.3	15.7
$> +0.15$	15.2%	84.5%	99.2	46.9	42.2	36.1	19.5
<i>perfect (truth) golden selection</i>			98.9	36.1	24.1	14.2	8.9

These diagonals are measured on a different sample from the analysis ones, and the two must not be mixed. The table above is built on the labelled *track* sample: every truth-matched pion track passing loose quality requirements (purity > 0.5 , track score > 0.5 , length > 5 cm), from events of any topology, 126 518 tracks in total. The migration diagonals quoted elsewhere in this note, 96/35/21/10/6% for five bins and 91/76% for two, are measured on the *analysis* sample: events passing the full $CC1\mu1\pi^\pm$ selection and satisfying the signal definition, about 10 700 per beam mode. The analysis sample has the better response, because

the selection removes exactly the events whose pion is poorly reconstructed. On the track sample the same five-bin scheme reads 99/24/13/6/6% and the two-bin scheme 98/62%.

Those track-sample figures are not the analysis response, and the 62% must not be read as the inclusive p_π measurement failing the usable-bin criterion. It does not: on the analysis sample the upper bin sits at 76%. The numbers in this table are correct for what they measure, and the conclusions drawn *within* it, which compare rows of the same sample, are unaffected.

The last row is the decisive one, and it must be read against the *efficiency* column rather than against the tightest cut. Perfect truth-level golden selection keeps 43.1% of tracks and gives 36.1, 24.1, 14.2, 8.9% in bins 2–5; the classifier at the matching 44.8% efficiency gives 33.1, 24.5, 15.4, 9.1%. They agree bin by bin. The classifier is therefore already performing as well as perfect knowledge of which pions stopped, and no better classifier exists to be built.

Multiple Coulomb scattering was tried as a pion momentum estimator, and as a classifier input. Range saturates because the pion interacts before stopping, which truncates the track. Multiple-Coulomb-scattering (MCS) momentum is inferred from the scattering along whatever track exists and is not truncated by an interaction, so it is the natural candidate to replace range. The CC0 π analysis uses exactly this pairing, range for contained tracks and MCS for exiting ones. Measured on 122 878 truth-matched pion tracks (`trk_mcs_muon_mom.v`, filled for 99.9% of them):

true p_π [GeV/c]	$\langle \text{len} \rangle$	$\langle \text{range/true} \rangle$	RMS	$\langle \text{MCS/true} \rangle$	RMS
0.175–0.250	25 cm	0.834	0.16	4.335	14.1
0.320–0.420	46 cm	0.629	0.23	2.438	7.2
0.550–0.800	71 cm	0.446	0.18	1.423	3.5
0.800–1.500	84 cm	0.319	0.16	1.011	2.1

The trade is stark. In the top bin range is biased to 0.319, which is the saturation, while MCS is essentially unbiased at 1.011; but MCS carries a 210% RMS there and 1400% in the lowest bin. It is unusable as a per-event estimator. The reason is structural and explains why the CC0 π recipe does not transfer: they apply MCS to *exiting* tracks, which are long by construction, whereas our problematic pions are contained but interacting, 25–84 cm long, and MCS needs of order a metre.

The same note uses the *disagreement* between the two estimators as a track-quality BDT input rather than as an estimator. That is well motivated here: a pion that interacts has a truncated range but an untruncated MCS, so the fractional difference $(p_{\text{range}} - p_{\text{MCS}})/p_{\text{range}}$ should flag precisely the events that break the response. Adding it to the golden-pion classifier, it ranks *second of ten* in importance, behind only the mean deflection. It nonetheless changes nothing:

classifier inputs	ROC integral	diagonals at $\sim 44\%$ efficiency
baseline (nine variables)	0.758	99.1/33.1/24.5/15.4/9.1
+ range–MCS difference	0.757	99.0/34.8/24.8/14.8/9.1

The interpretation is that the new variable is highly informative but not *additionally* informative: it is strongly correlated with the deflection variables already present, since both measure whether the track scattered heavily and ended abruptly. High importance rank with unchanged ROC is the signature of a redundant input. This is an independent confirmation of the ceiling argument above: the classifier already extracts what golden-ness has to offer, and the limit is the physics of the label rather than the quality of the classifier.

One tempting variant was deliberately not pursued. Retraining on the label “range is accurate”, $|p_{\text{corr}}/p_{\text{true}} - 1| < 0.2$, would target the quantity we actually care about and would certainly raise the diagonals. It would also be selecting on the answer: keeping events where

the estimator happens to be right sculpts the response by construction and inflates the diagonal without measuring anything, the same trap documented for the tight BDT working point above. The golden label is used precisely because it is a physical property of the pion, independent of how well it was reconstructed.

A caveat on the tighter working points. The > 0.15 row looks better than perfect truth selection (46.9, 42.2, 36.1, 19.5%), which is impossible for an honest classifier and is worth understanding rather than quoting. It arises because a hard cut on the classifier preferentially keeps *long* tracks: inside the top true bin, the surviving tracks have a mean length of 120.8 cm against 78.2 cm for all pions, and a mean reconstructed momentum of 0.406 against 0.307 GeV/ c . The cut is thus partly a cut on the reconstructed momentum itself, which raises the diagonal by construction while sculpting the spectrum. The apparent gain is circular and the tight working points should not be used. This is the same failure mode anticipated for track length as a direct input; it enters here through the deflection variables, which correlate with length, even though length itself was excluded from training.

Why, and what is left. Golden pions still reconstruct at $\langle p^{\text{corr}}/p^{\text{true}} \rangle = 0.73$ on this sample. Interaction is therefore not the only problem: the momentum conversion is itself biased. A single flat rescaling does *not* repair it, applying the best global factor drops the first bin from 98.9 to 49.1% and leaves 0/5 bins passing; because the bias is momentum-dependent, exactly as the saturation curve of Figure 1 shows.

Coarser binning helps but does not rescue it either. Collapsing to two bins (0.175, 0.230, ∞) gives 98.7 and 58.3% with perfect golden selection: better, still short of 0.68. Only a single bin; that is, no differential measurement at all, trivially passes.

Conclusion. The comparison is worth stating plainly rather than filing as a partial success. The classifier does what it was built to do and matches the published benchmark, but the p_π differential measurement above ~ 0.23 GeV/ c cannot be made to meet the BNB’s own binning criterion with a range-based estimator, with or without a golden-pion cut. The BNB analysis passes that criterion because its resolved bins (0.10–0.16, 0.16–0.19, 0.19–0.22) all sit *below* our 0.175 GeV/ c threshold, in the region where range still tracks momentum; above 0.22 it uses a single wide bin and an overflow. The realistic options for this analysis are therefore to lower the pion threshold toward 0.10 GeV/ c and resolve the region where the estimator works, to adopt a two-bin-plus-overflow p_π result and say so, or to replace the range-based estimator altogether. The golden-pion BDT is a useful ingredient in any of those, but it is not by itself the fix.

Scope of this conclusion. The failure described in this subsection concerns attempts to recover a finer multi-bin p_π spectrum, and its numbers are for the loose truth-matched track sample. It does not invalidate the adopted two-bin analysis-sample result (§1.1, Table 2), whose migration diagonals are 91% and 76%.

Scope of p_π . Both candidate routes to a finer p_π measurement were tried; as the preceding subsection and the following paragraph show, the golden-pion classifier does not repair the response even at perfect truth-level selection, whereas the proton tag does not change the response appreciably either way. At the two-bin scheme the inclusive selection already satisfies the binning criterion, so p_π need not be withdrawn from it; what it cannot support is the five-bin *shape* measurement originally planned. The other inclusive observables are unaffected: the angular variables and p_μ retain their shape information, and the total cross section does not use this estimator at all.

The proton-tagged route. The other candidate route to a sharper p_π response is the proton tag. It has been measured for FHC, RHC and the combined exposure: at the same two-bin edges the proton-tagged response is indistinguishable from the inclusive one, marginally worse rather than better. Applying the same migration-diagonal test gives

selection	config.	N_{sig} bin 1	bin 1 [0.175, 0.205]	bin 2 (> 0.205)	both > 0.68 ?
inclusive CC1mu1piXp	FHC	1056	91.0%	75.5%	yes
proton-tagged	FHC	469	88.9%	74.2%	yes
CC1mu1pi1p	RHC	621	89.4%	75.1%	yes
	comb.	1090	89.2%	74.7%	yes

measured on the numuMC overlays named in the corresponding `file_properties` lists, selected signal, with the reconstructed momentum mass-corrected exactly as the bin config does it. All four rows clear the criterion in both bins, and the proton-tagged values sit about two points *below* the inclusive ones rather than above them. The three proton-tagged configurations agree to better than a percentage point, so the response is a property of the selection rather than of a particular exposure, but the proton tag does not improve it.

That is the expected result. The pion-momentum response is set by whether the pion ranges out or interacts hadronically before stopping, which is a property of the pion's own passage through the argon. Requiring a reconstructed proton at the vertex changes the event sample, the purity and the efficiency, but it does not change what happens to the pion downstream, so there is no mechanism by which it would sharpen the momentum response, and the measurement says so directly.

The consequence is that the proton-tagged p_π measurement stands on its own merits, as the pion-momentum cross section of an exclusive final state that also yields W and TKI, and not as a repair of the inclusive one. The inclusive p_π at two bins is equally defensible on the response test.

Table 4: Two-bin p_π on the proton-tagged CC1mu1pi1p selection, all three beam configurations. $d\sigma/dp_\pi$ in $10^{-38} \text{ cm}^2/(\text{GeV}/c)/\text{Ar}$; bin 1 is [0.175, 0.205] GeV/c, bin 2 everything above. All curves are A_C -smeared. The proton-tagged `xsec` configs declare only the tune and the fake data, so no generator rows appear. Regenerated from the 2026-09-06 release (`report/tools/ppi2bin_tables.py`); the uncertainty row is the total.

	FHC		RHC		Combined	
	bin 1	bin 2	bin 1	bin 2	bin 1	bin 2
unfolded data	1.086	0.586	0.386	0.525	0.810	0.551
uncertainty	± 0.836	± 0.250	± 0.796	± 0.222	± 0.851	± 0.221
A_C truth	1.145	0.598	1.056	0.458	1.243	0.518
uB tune	1.139	0.591	1.135	0.469	1.287	0.521
unfolded/truth	0.95	0.98	0.37	1.15	0.65	1.06
σ_{int}	0.498		0.429		0.462	
χ^2/ndf	0.01/2 ($p = 1.00$)		1.21/2 ($p = 0.55$)		0.40/2 ($p = 0.82$)	

All three extractions close, with χ^2 of 0.01, 1.21 and 0.40 on two bins on the current release. The p_π differential cross section satisfies the field's binning criterion in every bin of every beam configuration here, as it does on the inclusive selection; the two are equivalent on that test, and the proton-tagged version is reported because it belongs to the exclusive final state, not because it is required to make p_π measurable.

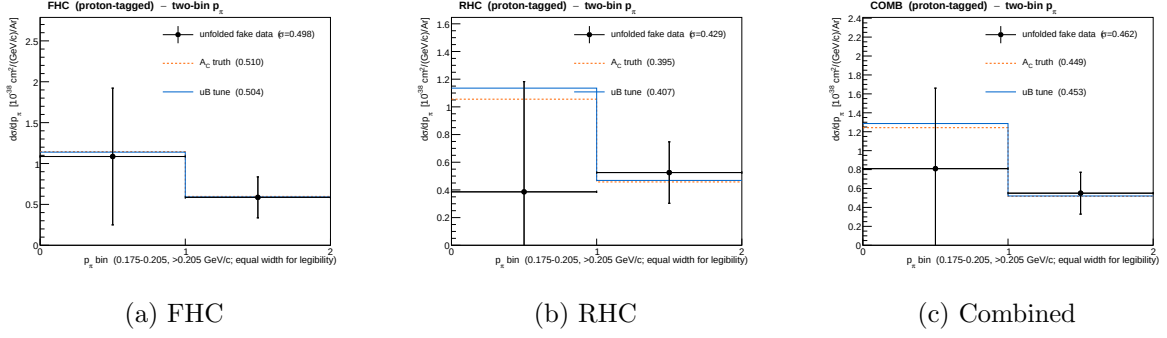


Figure 4: The two-bin p_π cross section on the proton-tagged CC1mu1pi1p selection, the graphical form of Tab. 4. Same edges as the inclusive result of Fig. 2 so the two are directly comparable, and the bins are again drawn equal-width.

The least-populated bin. The RHC first bin unfolds to 0.37 times its A_C -smeared truth (0.386 ± 0.796 against 1.056) with an unremarkable closure χ^2 of 1.21/2: a 0.8σ fluctuation on a bin whose uncertainty is twice its content. Independent throws move it by more than $\pm 1\sigma$. The bin holds too few events to say anything, and any real-data value in it should be read with its uncertainty and not its central value.

One caveat remains. The 0.205 GeV/c boundary was independently preferred by the migration-level edge scan run on the proton-tagged selection (`macros/ppi_binning_scan.C`, FHC; `logs/binning_scan/1p_fhc.txt`) as well as by the inclusive scan, so it is not merely inherited. It was not, however, re-optimised through a separate universe, covariance and unfolding production for every candidate boundary, which would need its own universe construction per configuration (~ 1 –2 h each); that is not expected to change the conclusion, since both bins clear the criterion with margin in all three proton-tagged configurations.

2 Rejected: extending the pion threshold below 0.175 GeV/c

The extension described here was tested and rejected; it is not part of the measurement, whose threshold is $p_\pi > 0.175$ GeV/c.

The low-momentum bin, as a test. The extension was implemented for p_π : a $[0.113, 0.175]$ GeV/c bin is prepended to the pion-momentum binning, lowering the measured phase space to the ν_e threshold. The truth-level signal below 0.175 GeV/c is recovered from the stored signal-component branches (the reconstruction reproduces MC_Signal above 0.175 GeV/c to 0.4%, the residual being the unstored kaon veto); 3323 true signal events populate the new bin. Its selection efficiency is 2.2%; an order of magnitude below the 15–17% of the standard bins (Table 5), as anticipated from the reconstruction turn-on. Despite this, the Wiener-SVD extraction remains well behaved with the bin included: the M_A^{RES} fake-data closure gives $\chi^2/\text{ndf} = 1.41/6$ ($p = 0.97$) and the flux term is unchanged (17% reco, 18.4% unfolded, amplification 1.08 $\times$). The bin therefore adds to the measurement, at the price of a large per-bin uncertainty driven by the low efficiency.

The generator predictions have been regenerated on the six-bin scheme (the ppi histogram uses $p_\pi > 0.113$ while the other observables retain $p_\pi > 0.175$, mirroring the analysis). GENIE, NuWro and NEUT all provide the low bin; GiBUU could not be re-binned as its generator-level event record was not retained, so it is omitted from the extended p_π comparison. Against the (fake) data the generators give $\chi^2/\text{ndf} = 11.1/6$ (GENIE, $p = 0.09$), 7.0/6 (NuWro, $p = 0.33$) and 6.3/6 (NEUT, $p = 0.39$): NEUT and NuWro describe the extended spectrum well, while GENIE is mildly disfavoured, consistent with the known GENIE over-prediction of low-momentum CC $\pi^\pm$

production.

Table 5: Selection efficiency per true p_π bin with the low-momentum bin added. The new $[0.113, 0.175]$ GeV/ c bin sits at 2.2%, an order of magnitude below the standard bins.

True p_π [GeV/ c]	N_{true}	N_{reco}	Efficiency
0.113–0.175 (new)	3323	74	2.2%
0.175–0.250	3368	526	15.6%
0.250–0.320	2190	367	16.8%
0.320–0.420	2831	479	16.9%
0.420–0.550	2499	410	16.4%
0.550–1.000	2933	429	14.6%

3 Proton-tagged selection, binning and additional smearing

The proton-tagged cut-flow, its per-observable binning study and its A_C factors are collected here rather than in the analysis note. The note reports the extracted proton-tagged cross sections and the reasons that sample is secondary scope; the material below is the supporting detail.

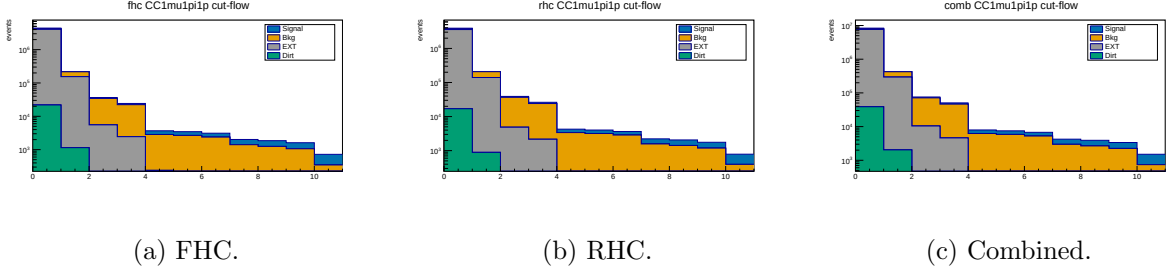


Figure 5: Proton-tagged (CC1mu1pi1p) cut-flow yields at full exposure (log scale; MC-prediction stack, Okabe-Ito: signal blue, background orange, EXT grey, dirt green). The last stage, IdentProton, is the leading-proton identification that defines the subsample.

Table 6: Proton-tagged (**CC1mu1pi1p**) cut-flow yields at full exposure for FHC, RHC and combined (MC prediction; data blind). Stages None–Nonprotons are the inherited inclusive selection with the proton-tagged signal definition; the final IdentProton stage adds the leading-proton identification. Signal/Bkg from the ν_μ overlays POT-scaled per run, reprocessed with the proton-tagged selection and the full central-value weight under the safe-weight rule (**reprocess_cutflow_cf_1p.sh**); EXT from the per-run beam-off samples scaled by the run gate ratios; dirt from the GENIE NuMI dirt overlay (**macros/cutflow_yields_1p.C**). The final row agrees with the proton-tagged cut-and-count check of the analysis note (Table 21 of the analysis note). Pred. = Signal + Bkg + EXT + Dirt. Eff. is the cumulative signal efficiency relative to the generated signal (the None stage, where all true signal is counted); Pur. = Signal/Pred.

Cut	Signal	Bkg MC	EXT	Dirt	Pred.	Eff. [%]	Pur. [%]
<i>FHC (Runs 1,2,4,5; 8.857×10^{20} POT)</i>							
None	3 295	270 438	4 049 292	21 864	4 344 889	100.0	0.08
InFiducialVol	2 763	61 976	153 533	1 146	219 418	83.9	1.3
Topological	2 183	28 864	5 409	125	36 580	66.2	6.0
MuonCandidate	1 977	19 865	2 363	84	24 288	60.0	8.1
ContainedPion	830	2 608	223	16	3 676	25.2	22.6
MuonIn3Planes	790	2 472	185	11	3 458	24.0	22.9
PionIn3Planes	733	2 230	144	4.4	3 112	22.3	23.6
ShowerCut	613	1 294	108	3.8	2 019	18.6	30.4
OpeningAngle	609	1 207	41	1.2	1 858	18.5	32.8
Nonprotons	550	1 031	30	0.8	1 612	16.7	34.1
IdentProton	376	338	8.5	0.3	723	11.4	52.1
<i>RHC (Runs 1,2,3,4a,4b,4c; 1.108×10^{21} POT)</i>							
None	3 427	288 891	3 560 975	16 958	3 870 250	100.0	0.09
InFiducialVol	2 886	67 416	139 122	889	210 312	84.2	1.4
Topological	2 318	31 870	4 767	97	39 052	67.7	5.9
MuonCandidate	2 111	21 962	2 090	65	26 227	61.6	8.0
ContainedPion	882	3 181	191	12	4 267	25.7	20.7
MuonIn3Planes	842	3 015	158	8.3	4 023	24.6	20.9
PionIn3Planes	784	2 739	121	3.4	3 648	22.9	21.5
ShowerCut	647	1 480	86	2.9	2 216	18.9	29.2
OpeningAngle	643	1 384	33	0.9	2 060	18.8	31.2
Nonprotons	573	1 162	23	0.6	1 759	16.7	32.6
IdentProton	391	388	7.1	0.2	786	11.4	49.8
<i>Combined FHC+RHC (1.994×10^{21} POT)</i>							
None	6 721	559 329	7 610 266	38 822	8 215 139	100.0	0.08
InFiducialVol	5 649	129 392	292 655	2 035	429 731	84.0	1.3
Topological	4 501	60 734	10 176	221	75 632	67.0	6.0
MuonCandidate	4 088	41 826	4 453	148	50 515	60.8	8.1
ContainedPion	1 712	5 790	414	28	7 943	25.5	21.6
MuonIn3Planes	1 632	5 487	343	19	7 481	24.3	21.8
PionIn3Planes	1 518	4 969	266	7.8	6 760	22.6	22.5
ShowerCut	1 260	2 774	194	6.7	4 235	18.7	29.7
OpeningAngle	1 251	2 591	74	2.2	3 918	18.6	31.9
Nonprotons	1 123	2 193	53	1.5	3 371	16.7	33.3
IdentProton	768	726	16	0.5	1 509	11.4	50.9

Table 7: Released binning of the eight W/TKI and angular proton-tagged observables, read from the data release (`curves_1p_*.tsv`). The two mass variables keep six equal-population bins chosen from the validated true-signal distributions, with the last bin open at truth level (the upper edge listed is the plotting boundary); the four TKI variables use the coarsened two-bin scheme adopted after the six-bin scheme was found to sit inside the resolution (§4.1). $W_{\pi p}$ brackets the $\Delta(1232)$ peak but its differential shape is withdrawn. The five inherited kinematic observables (p_μ , p_π , $\cos\theta_\mu$, $\cos\theta_\pi$, $\theta_{\mu\pi}$) use the inclusive-analysis binning of Appendix A of the analysis note unchanged.

Observable	Definition	Bins	Released bin edges
$W_{\pi p}$	$M(\pi p)$, direct	6 (withdrawn)	1.08, 1.19, 1.23, 1.27, 1.34, 1.47, 2.9 GeV/ c^2
W_{had}	calorimetric	6	0, 0.78, 1.08, 1.21, 1.31, 1.44, 2.74 GeV/ c^2
δp_T	$ \vec{p}_T^\mu + \vec{p}_T^{\text{had}} $	2	0, 0.3, 2.5 GeV/ c
$\delta\alpha_T$	TKI angle	2	0, 120, 180°
$\delta\phi_T$	TKI angle	2	0, 50, 180°
p_n	inferred nucleon momentum	2	0, 0.375, 2 GeV/ c
θ_p	proton polar angle	4 [†]	0, 0.659, 1.068, 1.539, π rad
$\theta_{\pi p}$	π - p opening angle	3 [†]	0, 1.193, 2.010, π rad

[†] Proposed secondary scope, added 2026-09-18; not part of the 2026-09-06 release. θ_p 's edges are derived on the RHC chain.

3.1 Additional smearing in the proton-tagged observables

Because the Wiener–SVD additional-smearing matrix A_C is not norm-preserving, each observable's predictions are rescaled by a different factor when they are brought into the measurement space. Table 8 gives that factor, measured as the ratio of the A_C -smeared to the truth-level generator integral, for the six proton-tagged observables. Its dominant behaviour is set by the response and the binning; the exact factor is prediction-dependent, and for the tested generators GENIE and GiBUU agree to a few per cent throughout.

On the current release the six proton-tagged observables are suppressed by similar amounts, 0.61–0.79 (Table 8): the two-bin TKI variables and $W_{\pi p}$ sit at 0.70–0.79 and the six-bin W_{had} , the most affected, at 0.61. Sharpness alone does not drive the suppression: $W_{\pi p}$ is the sharpest distribution measured here (the ratio of its largest to smallest bin is ~ 36) yet is suppressed no more than the TKI variables, because A_C removes structure that the *response* cannot resolve and $W_{\pi p}$ is built from directly measured pion and proton momenta with equal-population bins that track the $\Delta(1232)$ peak. No observable is enhanced by A_C (a physical smearing cannot amplify; an amplification would signal a mis-specified regularisation, §5.1). The suppression is applied identically to data, tune and generators, so model comparisons read off any panel are made on equal footing.

Table 8: Mean A_C row sum for the proton-tagged observables (FHC), from the released matrices `A_C_1p_FHC5_*.tsv`, release 2026-09-06; equal to the ratio of the A_C -smeared to the truth-level integral for a flat prediction. The inclusive observables are given for scale.

[†] From the 2026-09-18 extraction rather than the 2026-09-06 data release, which does not contain these

	Observable	mean A_C row sum
observables.	p_n	0.79
	$\theta_p^\dagger$	0.73
	$\theta_{\pi p}^\dagger$	0.61
	$\delta\alpha_T$	0.79
	$W_{\pi p}$	0.75
	$\delta\phi_T$	0.72
	δp_T	0.70
	W_{had}	0.61
<i>Computed from the released matrices (<code>A_C_1p_FHC5_*.tsv</code>). The values span 0.61–0.79, a range of 0.18, so differences in smearing cannot order the generator χ^2 across observables.</i>		

4 Hadronic mass, model comparison, and estimator surveys

4.1 The W binnings are finer than the resolution

$W_{\pi p}$ shows the least model tension of the proton-tagged observables. The cause is not physics.

The bin edges were chosen for equal population, six bins of roughly equal content, without reference to the resolution. Measured on selected signal, FHC Runs 1/2/4:

	$W_{\pi p}$	W_{had}
residual RMS [GeV/ c^2]	0.142	0.283
robust half-width (16–84%)	0.101	0.177
bias (mean reco – true)	−0.111	−0.032
narrowest analysis bin	0.040	0.100
widest analysis bin	1.430	1.300
width ratio	35.7	13.0

The second and third $W_{\pi p}$ bins are 40 MeV/ c^2 wide against a resolution of 101–142 MeV/ c^2 : they are two to three times narrower than the detector can resolve. The bias alone, −111 MeV/ c^2 , is larger than three of the six bin widths. The migration diagonals follow directly, 87.6/22.3/11.7/12.0/10.0/2.3%: only the first bin is measured, and the top bin at 2.3% is essentially unmeasured.

This is the same failure as the five-bin p_π scheme, and for the same underlying reason. $W_{\pi p}$ is reconstructed from the pion and proton momenta, so it inherits the saturation of the range-based pion momentum described in §1.1; its resolution cannot be better than that of its inputs.

An exhaustive edge scan, the same procedure and criterion used for p_π , gives:

scheme	diagonals	verdict
current, six bins	87.6/22.3/11.7/12.0/10.0/2.3%	1 of 6 usable
best two bins (split 1.15)	83.2/77.9%	both usable
best three bins	83.2/49.7/45.9%	fails
best four bins	83.2/45.0/28.2/28.2%	fails

As with p_π , two bins is not merely the best option but the only one that satisfies the criterion, and the first bin is thin (137 selected signal events, against a floor of 100). The implication for the model comparison is direct: a low generator χ^2 at six bins would reflect a binning that cannot resolve the differences between generators, not agreement between them. The same test should be applied to W_{had} , whose upper bins are also narrower than its 177 MeV/ c^2 half-width, before either W result is quoted as a shape measurement. *[Open: the W_{had} edge scan has not been run; W_{had} is released at six bins with its closure and generator overlays, and its shape should be read with this caveat (analysis note, Limitations).]*

Adopted scheme: two bins split at 1.15 GeV/ c^2 . The boundary is more tightly constrained than it was for p_π , where anything between roughly 0.19 and 0.22 worked. Here only one choice satisfies both the diagonal criterion and the 100-event floor:

boundary [GeV/ c^2]	bin 1	bin 2	$N(\text{bin } 1)$	
1.13	76.5%	88.0%	51	below the event floor
1.15	83.2%	77.9%	137	adopted
1.17	86.8%	66.8%	266	bin 2 just fails
1.19	87.6%	56.0%	468	fails
1.23	91.7%	37.6%	1114	fails

Moving the boundary up buys population in the first bin at the direct expense of the second: the two diagonals trade against each other steeply, and the window in which both exceed 0.68 is only about ± 0.01 GeV/ c^2 wide. That narrowness is itself a consequence of the resolution being comparable to the full width of the peak region.

Two consequences should be carried with the result. First, **the first bin is thin**, 137 selected signal events against a floor of 100, so its statistical uncertainty will be large once systematics are folded in; the p_π first bin, with 469 events, carries 77% (Table 4). Second, if that proves too sparse in practice, 1.17 is the natural fallback: it doubles the population for a bin-2 diagonal of 66.8%, marginally below threshold, which is a defensible trade to state explicitly rather than a criterion to fail silently.

The scheme is implemented in `configs/ccpi1pWpibr_bin_config_2bin.txt`, with the reasoning recorded alongside it in a companion `.README`. Running it through the extraction settled the question (§4.2): the two-bin scheme meets the migration criterion but leaves only about 137 selected signal events in the first bin, which the Wiener filter then suppresses, driving A_C towards rank one for statistical rather than algebraic reasons. Widening the first bin restores its population at the cost of the upper-bin diagonal, and no scanned edge satisfies both requirements at once.

The apparent $W_{\pi p}$ agreement therefore does *not* survive a binning the detector can resolve, and the differential shape is withdrawn: at this exposure the proton-tagged sample supports an integrated cross section. W_{had} carries the same limitation, since it depends on the same pion momentum.

4.2 What limits $W_{\pi p}$, and what can be done about it

The two-bin $W_{\pi p}$ scheme of §4.1 was built and run. It does not work, for a reason distinct from the six-bin failure, and the investigation that followed settles what the pion momentum can and cannot deliver.

The two-bin extraction fails on statistics. The *response* behaves exactly as designed: the framework returns migration diagonals of 0.802 and 0.773, matching the truth-level scan, and the closure is acceptable at $\chi^2 = 1.17/2$ ($p = 0.56$). The *extraction* nonetheless carries almost no shape information, because A_C collapses:

(Matrices from the pass in which the two-bin $W_{\pi p}$ scheme was run, written with true bins as rows; the two p_π rows are the same-pass comparison, and the released inclusive p_π matrix is that of Fig. 3.)

two-bin scheme	A_C	s_2/s_1
$W_{\pi p}$, first bin $N = 137$	$\begin{pmatrix} 0.018 & -0.410 \\ 0.034 & 0.892 \end{pmatrix}$	0.031
p_π proton-tagged, $N = 469$	$\begin{pmatrix} 0.498 & -0.089 \\ 0.013 & 0.854 \end{pmatrix}$	0.577
p_π inclusive, $N = 1056$	$\begin{pmatrix} 0.594 & -0.382 \\ 0.011 & 0.891 \end{pmatrix}$	0.528

The top-left entry is 0.018: A_C passes under two per cent of the first bin and takes -0.41 from the second. This is a rank-one collapse of the kind described in §5.1, but its cause is statistical rather than a defective regulariser. With 137 events the first bin has such poor signal-to-noise that the Wiener filter suppresses that mode almost entirely, and the ordering across the three schemes, $137 \rightarrow 0.031$, $469 \rightarrow 0.577$, $1056 \rightarrow 0.528$, follows population rather than anything geometric.

$W_{\pi p}$ is therefore caught between two requirements. Bins narrow enough to satisfy the response criterion leave the first bin too sparse to unfold; bins wide enough for a stable unfold (1.19 gives 468 events, matching the p_π case that works) fail the criterion at 56%.

No binning escapes it. The conclusion does not rest on the one scheme that was tried. Scanning *every* two- and three-bin scheme with edges between 1.12 and 1.70 GeV/ c^2 (FHC Run1+2+4, 3505 selected signal events) finds none that keeps both a migration diagonal above 0.68 and adequate statistics in every bin:

two-bin edge	N_1/N_2	diag ₁ / diag ₂	
1.15	137/3367	0.832/0.779	first bin too sparse
1.18	351/3153	0.866/0.611	
1.24	1298/2206	0.930/0.347	
1.27 (median)	1744/1760	0.951/ 0.261	upper bin unusable
1.45	2966/538	0.993/0.030	

The trend exposes the mechanism. Reconstructed $W_{\pi p}$ is biased about 111 MeV low, inheriting the saturation of the range-based pion momentum (§1.1), so raising the edge far enough to balance the statistics drains the upper bin by downward migration: diag₁ climbs from 0.786 to 0.999 across the scan while diag₂ collapses from 0.927 to 0.004. There is no edge at which both bins are simultaneously populated and diagonal, and no three-bin combination does better — the third-bin diagonal never exceeds 0.347.

At the current exposure $W_{\pi p}$ does not support a shape measurement and is quoted as an integrated cross section unless the pion momentum improves. That integrated number is a one-bin extraction rather than a regularised unfolding: with a single bin there is nothing to unfold, since the response is 1×1 , A_C is trivially unity and the regularisation has no effect. It is reported through the total flux-averaged cross-section path of Sec. 11.3 of the analysis note, and integrated over the full range it is the same quantity as the total proton-tagged cross section. The corresponding integrated generator predictions are built by `generator_predictions/newg4/make_wpipr_1bin.C`, summing the per-bin cross sections of the six-bin flux-averaged predictions.

Six-bin $W_{\pi p}$ closure, diagnostic only. For completeness the six-bin extraction was carried through the full chain in all three configurations (detector variations included). It closes against the A_C -smeared truth, but that closure cannot restore shape information the response does not contain (the top bin is 2% diagonal), so these numbers are a check of the machinery and *not* a result, and they do not appear in the analysis note’s result tables. The six-bin generator

predictions (GENIE 0.501, NuWro 0.646, GiBUU 0.735, NEUT 0.783×10^{-38} cm²/Ar, FHC), exist in the release for the same diagnostic purpose.

Table 9: Six-bin $W_{\pi p}$ fake-data closure, diagnostic only (withdrawn as a differential result). σ_{int} in 10^{-38} cm²/Ar; columns as in the note’s proton-tagged closure tables.

Config	σ_{int}	unf./truth	χ^2/ndf
FHC	0.510	1.06	0.52/6
RHC	0.305	0.88	2.56/6
Combined	0.390	0.92	1.60/6

The limitation is the pion momentum, and only the pion momentum. Correlating the $W_{\pi p}$ residual against each input residual isolates the cause:

input	r	W residual RMS when...	
pion momentum	+0.726	all events	0.142
proton momentum	+0.012	pion momentum well reconstructed	0.061
pion $\cos \theta$	−0.029	proton momentum well reconstructed	0.130
proton $\cos \theta$	−0.032	both well reconstructed	0.038

Fixing the pion momentum alone takes the resolution from 0.142 to 0.061, a factor 2.3; fixing the proton buys almost nothing, and the angles are irrelevant. A single defect propagates through the whole analysis: the hadronic interaction that truncates the pion track forces p_π to two bins, and forces $W_{\pi p}$ out of a shape measurement altogether.

Reparameterising to W^2 does not help. A monotonic transformation with mapped edges leaves the migration matrix unchanged, which was verified directly: splitting at 1.15 GeV/ c^2 in W and at $1.15^2 = 1.3225$ GeV²/ c^4 in W^2 gives diagonals of 83.2% and 77.9% in both cases, on identical events. Resolution and bin width scale together, $\sigma(W^2) = 2W\sigma(W)$, so σ/width is invariant at about 2 either way. This is the same lesson as θ_μ versus $\cos \theta_\mu$ (§5.1).

4.3 Survey of pion momentum estimators without a magnetic field

Four approaches were measured on truth-matched pion tracks. All are limited by the same mechanism, that each measures what the pion did *before* it interacted:

estimator	$\langle \text{reco}/\text{true} \rangle$, top bin	RMS	note
range (analysis)	0.319	0.16	saturates near 0.34 GeV/ c
calorimetric energy	0.383	0.31	also truncated, and $2\times$ noisier
MCS	1.011	2.1	unbiased, unusably noisy on short tracks
golden-pion selection	0.327	0.18	no gain above 0.8 GeV/ c

Calorimetry does not escape the truncation. Deposited charge stops at the interaction point just as range does, so $\langle \text{calo}/\text{true} \rangle$ falls from 1.136 to 0.383 across the momentum range while the RMS is three to five times that of range (1.007 to 0.313, against 0.157 to 0.158). It is the same measurement with more noise.

The golden-pion selection does not help where it is needed. Splitting the sample by the truth label shows that golden pions are not well reconstructed either, and that their advantage disappears exactly where it is needed:

true p_π [GeV/ c]	golden fraction	golden $\langle \text{range/true} \rangle$	non-golden
0.175–0.250	77.8%	0.873	0.699
0.320–0.420	44.1%	0.705	0.570
0.550–0.800	21.4%	0.507	0.429
0.800–1.500	17.2%	0.327	0.317

Above 0.8 GeV/ c the golden and non-golden values are identical to within one per cent. The reason is physical: a 1 GeV/ c pion has a CSDA range of several metres in argon and cannot stop inside a 2.5 m TPC unless it has already lost most of its energy through processes that truncate or break the track. “Golden” at high momentum therefore selects a pathological subsample rather than a clean one, which is why the classifier of §1.1 reaches a ceiling: above ~ 0.5 GeV/ c there is no clean subsample to find.

A multivariate regression fixes the bias but redistributes the resolution. A BDTG regression on p_π^{true} , using range, MCS, calorimetry, length, the three deflection variables, the four Bragg likelihoods, track score and daughter count, was trained on Runs 1 and 4 and evaluated on the held-out Run 2. It largely removes the saturation in the mean, $\langle \text{pred/true} \rangle$ running 1.196 to 0.722 against 0.835 to 0.319 for range, at comparable RMS. Rebuilding $W_{\pi p}$ with it:

pion estimator	W bias	W RMS
range (analysis)	−0.127	0.144
BDTG regression	− 0.005	0.119
true p_π (ceiling)	+0.003	0.046

The bias falls by a factor of 25, which matters independently of binning: a -127 MeV/ c^2 shift distorts the whole W distribution. The resolution improves only from 0.144 to 0.119 against a ceiling of 0.046, so the regression recovers about a quarter of what is available.

Its effect on the binning is a redistribution rather than a rescue. The six-bin diagonals move from 98/30/19/19/13/6% to 63/38/33/43/52/56%: the top bin improves nearly tenfold, but the first bin degrades from 98% to 63%, and in the two-bin scheme the worst bin falls from 73% to 58%. That degradation is regression to the mean, the estimator being pulled toward the sample centre, and it damages the low- W region where the resonance physics sits. The same pattern appears in p_π itself, where the first bin falls from 99.2% to 60.6% while the top bin rises from 0.5% to 33.2%.

Two objections stand against adopting it. The estimator would be MC-trained and sits upstream of everything, so its model dependence becomes a systematic on every result rather than on one cut. And the trade it offers moves measurability out of the region of interest. It is not adopted here.

What would actually work. The ceiling measurements above bound what is available: a perfect pion momentum would give $W_{\pi p}$ a resolution of 0.046, a factor of three better than the regression achieves and enough to support several bins. The information is carried away by the interaction products, so the approach that attacks the cause is to recover their energy: the pion and its daughters treated as a mini-calorimeter, using the per-PFP calorimetric energies and daughter links already present in the input files. For the proton-tagged sample there is a second, independent handle in transverse kinematic imbalance, where the muon and proton constrain the pion transverse momentum without using the pion’s own reconstruction at all, at the cost of Fermi-motion smearing of order 250 MeV/ c . The transverse-imbalance route has not been tried; the daughter-recovery route was tested next.

Recovering the interaction products does not work either. The ceiling measurements above locate the missing information in the secondaries produced when the pion interacts, so the natural remedy is to treat the pion and its products as a single calorimeter. This was tested. PeLEE carries no parent index, so association is geometric, by PFP start point relative to the pion track end, optionally restricted to the generation immediately below the pion in the Pandora hierarchy. The hierarchy restriction matters: at a 40 cm radius the unrestricted sum collects 0.371 GeV in the lowest momentum bin, where only 0.034 GeV is missing, so it is dominated by unrelated activity.

With the hierarchy cut applied, on Run 1:

true p_π [GeV/c]	$\langle \text{KE missing} \rangle$	$\langle \sum E \text{ of daughters within } R \rangle$ [GeV]			
		5 cm	10 cm	20 cm	40 cm
0.175–0.420	0.034	0.023	0.026	0.029	0.037
0.420–0.800	0.246	0.040	0.048	0.055	0.062
0.800–1.500	0.659	0.056	0.074	0.091	0.104

“KE missing” is the pion’s true kinetic energy less the calorimetric energy of its own track. In the top momentum bin 659 MeV is unaccounted for and the visible daughters supply 104 MeV, of which about 37 MeV is the baseline seen even in the lowest bin where almost nothing is missing. Genuine recovery is therefore of order 10% at high momentum and 25% in the middle bin.

It also does not change the binning assessment, which was tested directly rather than inferred from the bias. Adding the daughter energy improves $\langle \text{reco/true} \rangle$ everywhere, the top bin moving from 0.373 to 0.470 and the first bin becoming nearly unbiased at 0.953, and it roughly triples the upper migration diagonals. Neither is enough: the best upper diagonal rises from 5.5% to 15.7%, against the 68% required. In the adopted two-bin scheme it is a wash and arguably a loss, the worst bin improving from 61.4% to 71.0% while the first degrades from 97.7% to 82.4%, the same regression-to-the-mean behaviour seen with the multivariate estimator. Since both bins already clear the criterion with plain range, nothing is rescued and the daughter association would be imported as a new systematic for no gain. **The two-bin choice for p_π is therefore robust against every estimator tested**, which is a stronger statement than the binning study alone could make.

That recovery is far too little to change the measurement, and the reason is physical rather than algorithmic. Pion absorption on argon, $\pi A \rightarrow NN \dots$, deposits much of the energy in nucleons, roughly half of which are neutrons and therefore invisible in liquid argon; the charged products are frequently short and fall below the threshold to be reconstructed as separate particles. The energy is not merely unassociated, it is largely undetected. Applying the daughter sum would move $\langle \text{reco/true} \rangle$ in the top bin from 0.32 to about 0.42, still less than half of truth.

Where this leaves the pion momentum. Five approaches were measured: range, calorimetry, MCS, golden-pion tagging and a multivariate regression, together with recovery of the interaction products. Each is limited by the same fact, that a charged pion which interacts before stopping destroys the information a non-magnetised detector uses to infer its momentum, and that a substantial part of the energy leaves the detector unseen. The regression is the only one that materially improves anything, removing the $W_{\pi p}$ bias, and it does so at the cost of model dependence and of the low- W region.

One avenue remains untried and is independent of the pion’s own reconstruction: for the proton-tagged sample, transverse kinematic imbalance constrains the pion transverse momentum from the muon and proton alone. It is limited by Fermi motion, of order 250 MeV/c, which is comparable to the present errors rather than better than them, so it is worth measuring but should not be expected to transform the measurement. Failing that, the honest conclusion is that p_π is a two-bin measurement and $W_{\pi p}$ an integrated one at this exposure, and that both are limited by physics rather than by analysis choices.

4.4 Cross-check methods

Provenance. The D’Agostini cross-check (Table 11 and the numbers around it) is evaluated on the release. The closure figures of the other methods were evaluated on an earlier fake-data throw; the comparison they support is a property of the operators and does not depend on the throw.

Seven additional methods are run in parallel:

Table 10: Unfolding methods.

Label	Description
<code>wiener_svd</code>	Wiener-SVD (primary result)
<code>bin_by_bin</code>	Efficiency correction only; no migration unfolding
<code>ibu</code>	D’Agostini iterative Bayesian, 4 iterations
<code>tikhonov</code>	Tikhonov-regularised least-squares, $\tau = 1.0$
<code>roo_bayes</code>	RooUnfoldBayes, 4 iterations
<code>roo_svd</code>	RooUnfoldSvd
<code>roo_invert</code>	RooUnfoldInvert (direct matrix inversion)
<code>roo_binbybin</code>	RooUnfoldBinByBin

The D’Agostini iterative-Bayesian method provides the principal quantitative cross-check of the Wiener-SVD result. Because it does not apply the Wiener-SVD additional-smearing matrix A_C , its unfolded total is free of the (non-norm-preserving) A_C suppression and is stable against the iteration count. Table 11 compares the two integrated totals for all three configurations: the D’Agostini result sits above the Wiener-SVD total by 31% (FHC), 29% (RHC) and 23% (Combined), equivalently A_C suppresses the quoted Wiener-SVD integral by 23%, 23% and 19%, and it varies by $\leq 0.3\%$ across 1, 2 and 4 iterations. The D’Agostini integral is flat across the five observables to 1.2% (FHC, 1.017–1.028), 0.9% (RHC, 0.930–0.939) and 0.5% (Combined, 0.972–0.976), against Wiener-SVD spreads of 24%, 35% and 34%, directly confirming that the observable-to-observable spread in the Wiener-SVD integrated cross sections (Table 20 of the analysis note) is the A_C smearing rather than a physical or acceptance effect.

Table 11: Wiener-SVD vs. D’Agostini (iterative Bayesian, 4 iterations) integrated cross section, averaged over the five inclusive observables (p_μ , two-bin p_π , $\cos\theta_\mu$, $\cos\theta_\pi$, $\theta_{\mu\pi}$; 10^{-38} cm²/Ar), on the 2026-09-06 release. D’Agostini carries no A_C smearing, so it recovers a 23–31% higher total, and it is flat across observables (spread 1.2% FHC, 0.9% RHC, 0.5% combined) where the Wiener-SVD integrals spread by 24–35%, confirming that the observable-to-observable variation of σ_{int} in Table 20 of the analysis note is the A_C smearing rather than a physical or normalisation effect. The last column is the change of the D’Agostini integral between 1 and 4 iterations.

Config	$\langle\sigma\rangle_{\text{WSVD}}$	$\langle\sigma\rangle_{\text{DAG}}$	DAG/WSVD	DAG 1→4 iter
FHC	0.782	1.022	1.31	$\leq 0.3\%$
RHC	0.723	0.933	1.29	$\leq 0.3\%$
Combined	0.790	0.974	1.23	$\leq 0.1\%$

5 Angular observables: the $\cos\theta_\mu$ versus θ_μ forensics

5.1 Angular variable: $\cos\theta_\mu$ versus θ_μ

The muon-angle distribution is measured in both $\cos\theta_\mu$ and θ_μ . The two span the same full range but are different partitions of it: the θ_μ edges (0.43, 0.62, 0.84, 1.17 rad) are not the arccos

images of the $\cos\theta_\mu$ edges (0.45, 0.64, 0.86, 1.10 rad; Table 15 of the analysis note), and they behave very differently under the Wiener-SVD additional smearing A_C , and the choice matters for how the measurement compares to models.

In $\cos\theta_\mu$ the signal is sharply forward-peaked: the last two bins ($\cos\theta_\mu > 0.8$) hold most of the rate, and A_C suppresses the integral of any truth-level prediction strongly (row sums 0.17–0.39 in FHC on the current release, against 0.66–0.91 for p_μ). Because the same A_C is applied to the data, the tune and every generator, they are compared on equal footing.

The suppression is not a property of the shape of the truth distribution. Across the observables the A_C integral factor shows no correlation with peakedness: $W_{\pi p}$ is the sharpest distribution in the analysis (peak-to-minimum ratio 36) yet its factor is 0.75, δp_T is comparatively flat (ratio 10) at 0.70, and the six-bin W_{had} , neither sharp nor flat, is the most suppressed at 0.61 (Table 8); the inclusive $\cos\theta_\mu$ factors sit far below all of these (0.17–0.39 against 0.66–0.91 for p_μ) because its response is the worst conditioned, not because its peak is the sharpest. Within $\cos\theta_\mu$ itself NEUT and NuWro are the *flattest* of the five truth-level curves (peak-to-minimum 3.8 and 4.3, against 16.3 for the tune) and are smeared no differently from the others: on the release no smeared generator bin is negative and both sit above the tune in every bin. A_C is the Wiener filter acting on the SVD modes of the response: it removes whatever the data cannot statistically constrain, which is set by the conditioning of the response for that observable and binning, not by the shape of the truth distribution, and it is not norm-preserving, so it can suppress or enhance.

Why the regularisation operator must respect the bin widths

Wiener-SVD regularises with a matrix C ; this analysis uses the second-derivative form, built on the *density* with the actual bin widths. The reason is best shown by what the plain tridiagonal stencil $(1, -2, 1)$ does on these binnings. That stencil is a second derivative only on a uniform grid; the released binnings are strongly non-uniform, with width ratios up to 14:1 (Table 13), so on them it is not a second derivative at all. It is near-singular, its row sums are the 10^{-6} regularisation floor, and its near-null direction is the constant vector rather than anything physical. The Wiener filter is then free to retain that direction alone, which is a rank-1 A_C that passes normalisation and discards shape.

The FHC θ_μ extraction is the clearest case, because its bin widths span a factor of 10.4. Unfolded with the uniform stencil, the unfolded data, the A_C -smeared truth, the MicroBooNE tune and all four generators share the same normalised shape to four decimal places (Table 12): writing the four generator predictions as the columns of an input matrix and their A_C -smeared counterparts as the columns of the output matrix, the inputs are full rank ($s_2/s_1 = 5.0 \times 10^{-2}$) while the outputs give $s_2/s_1 = 2.8 \times 10^{-6}$. Every shape degree of freedom is annihilated, each model is mapped onto one common curve and only its normalisation survives; any agreement with the data is then true by construction, and a model comparison made on such an extraction is meaningless. $\cos\theta_\mu$, the next most non-uniform binning at 14.5, is the other observable strongly affected; the remaining observables sit at width ratios of 3–6 and are affected far less.

Table 12: FHC θ_μ unfolded with a *uniform-grid* second-derivative stencil: normalised bin contents. Five independent inputs, the measurement and four generator predictions plus the tune, collapse onto one shape, the signature of a rank-1 A_C .

	bin 1	bin 2	bin 3	bin 4	bin 5
Unfolded data	0.1485	0.3361	0.2899	0.1934	0.0321
A_C -smeared truth	0.1484	0.3359	0.2901	0.1933	0.0323
MicroBooNE tune	0.1484	0.3359	0.2901	0.1934	0.0322
NEUT	0.1484	0.3359	0.2901	0.1934	0.0322

The operator used. Both derivative regularisation matrices difference the density on the actual grid: each column is divided by its bin width and the stencil uses the true spacings between bin centres, with a one-sided first-derivative stencil at the ends. (A Dirichlet closure was rejected: it asserts the distribution vanishes at the edges, which is false here, $\cos\theta_\mu$ peaks at its upper edge.) With no widths supplied the code reduces to the uniform stencil. The widths are passed in by the unfolding application, which holds the slice definitions that the cross-section extractor does not. The effect on the conditioning of A_C , measured as s_2/s_1 of the smearing matrix itself, is given in Table 13: every inclusive observable improves and none is degraded except combined $\cos\theta_\mu$, marginally; in the proton-tagged family combined $W_{\pi p}$ (0.022) and $\delta\phi_T$ (0.083) would be as compromised as FHC θ_μ under the uniform stencil.

Table 13: A_C conditioning s_2/s_1 with the uniform-grid stencil (“uniform”) and with the width-aware operator (“width-aware”), measured on one fake-data pass at the binnings of that time (the $\cos\theta_\pi$ row is a five-bin scheme since replaced); the width-aware values of the release follow from the released matrices. With the bin-width ratio of each observable. A value approaching zero means the smearing matrix has collapsed onto a single mode and the extraction carries no shape information.

observable	width ratio	FHC		RHC		COMB
		uniform	width-aware	uniform	width-aware	uniform $\rightarrow$ width-aware
p_μ	6.3	0.34	0.90	0.31	0.94	0.19 $\rightarrow$ 0.68
$\cos\theta_\mu$	14.5	0.46	0.57	0.45	0.55	0.42 $\rightarrow$ 0.39
$\cos\theta_\pi$	3.6	0.67	0.71	0.55	0.79	0.55 $\rightarrow$ 0.74
$\theta_{\mu\pi}$	3.0	0.42	0.85	0.47	0.65	0.42 $\rightarrow$ 0.46
θ_μ	10.4	0.015	0.51	0.22	0.62	0.045 $\rightarrow$ 0.38
p_π (2 bins)	—	0.46	0.53	0.41	0.44	0.29 $\rightarrow$ 0.40
<i>proton-tagged, combined configuration</i>						
$W_{\pi p}$	—					0.022 $\rightarrow$ 0.60
W_{had}	—					0.23 $\rightarrow$ 0.86
δp_T	—					0.25 $\rightarrow$ 0.93
$\delta\alpha_T$	—					0.24 $\rightarrow$ 0.48
$\delta\phi_T$	—					0.083 $\rightarrow$ 0.85
p_n	—					0.21 $\rightarrow$ 0.88

θ_μ on the release. The fake data are Poisson throws from the MicroBooNE tune, so the tune is the model that *ought* to fit best, and it does (FHC θ_μ , χ^2 on five bins: tune 2.14, $p = 0.83$; GENIE 5.55; GiBUU 6.07; NEUT 8.46; NuWro 9.10; closure 1.52, $p = 0.91$). Under the uniform stencil every model gave 2.8–4.7, indistinguishable, because every model had been mapped onto the same curve. Both parameterisations of the muon angle are released and should be read together; the θ_μ χ^2 are systematically the larger (analysis note, Table 15 of the analysis note).

5.2 Total flux-averaged cross section and systematic breakdown

The measurement is the one-bin extraction of Sec. 11.3 of the analysis note on the Poisson-thrown fake data (one true bin over the full $\cos\theta_\mu$ range, no overflow, Wiener filter off, $A_C \equiv 1$; `data.release/total_xsec.tsv`), with its full universe-propagated uncertainty; at unblinding it becomes the real beam-on extraction. It agrees with all four generators within 0.8σ : standalone GENIE sits 0.5–0.8 σ low, while GiBUU, NEUT and NuWro lie within 0.8σ .

Table 14: Total flux-averaged cross section from the one-bin extraction (Sec. 11.3 of the analysis note; $10^{-38} \text{ cm}^2/\text{Ar}$), compared with the flux-averaged generator predictions. The measurement uncertainty is the complete configured one-bin covariance (flux, cross section, re-interaction, detector, POT, targets and statistics; MCS-momentum, beamline-geometry, beam-off gate-ratio and Run-2 stand-in terms are not propagated, Sec. 15 of the analysis note). Release 2026-09-06, `data_release/total_xsec.tsv`. Values in parentheses give each generator’s deviation from the measurement in units of the measurement uncertainty. The generator predictions are integrated over the full binning with open upper bins (a closed upper edge on p_μ or p_π would discard 4–8% of each prediction).

Config	Measurement	GENIE	GiBUU	NEUT	NuWro
FHC	1.03 ± 0.40	$0.82 (+0.5\sigma)$	$1.13 (-0.3\sigma)$	$1.32 (-0.7\sigma)$	$1.25 (-0.6\sigma)$
RHC	0.93 ± 0.35	$0.67 (+0.8\sigma)$	$0.94 (-0.0\sigma)$	$1.10 (-0.5\sigma)$	$1.04 (-0.3\sigma)$
Combined	0.97 ± 0.37	$0.74 (+0.6\sigma)$	$1.03 (-0.1\sigma)$	$1.20 (-0.6\sigma)$	$1.14 (-0.4\sigma)$

The systematic breakdown by source (bin-averaged fractional uncertainty in cross-section space) is given for all three configurations in Tables 15–17. The flux (PPFX) term dominates throughout, followed by the detector variations; the data-statistical term reflects the fake-data POT of each configuration. The RHC detector term is the largest single-configuration entry, reaching 34.7% on p_π and 35.0% on $\cos \theta_\mu$.

Table 15: FHC ({Run 1,2,4,5}) systematic breakdown by source, bin-averaged fractional uncertainty (%), regenerated from the systematics dumps of the 2026-09-06 release (`logs/systdump`); “Prediction total” excludes the data statistics.

Source	p_μ	p_π	$\cos \theta_\mu$	$\cos \theta_\pi$	$\theta_{\mu\pi}$	θ_μ
Prediction total	40.7	50.2	42.0	38.8	36.5	31.0
Cross section (GENIE)	10.7	16.9	10.3	11.3	10.5	8.5
Flux (PPFX)	28.0	32.5	28.0	27.0	28.3	24.7
Detector	24.7	30.3	28.3	22.9	19.6	15.6
Reinteraction	4.4	11.9	3.6	5.2	4.0	3.5
MC stat	4.9	6.3	5.3	4.3	2.7	2.9
EXT stat	0.9	3.1	2.3	1.5	1.2	1.4
Data stat	14.3	18.3	15.4	12.7	8.2	8.6
POT + targets	3.4	4.2	3.6	3.4	3.5	3.0
Total (incl. data stat)	43.5	53.9	44.9	41.0	37.5	32.4

Table 16: RHC ({Run 1,2,3,4a,4b,4c}) systematic breakdown by source, bin-averaged fractional uncertainty (%); same provenance as Table 15.

Source	p_μ	p_π	$\cos \theta_\mu$	$\cos \theta_\pi$	$\theta_{\mu\pi}$	θ_μ
Prediction total	37.7	48.6	46.3	38.5	32.6	36.1
Cross section (GENIE)	12.0	14.4	9.1	10.8	10.7	10.7
Flux (PPFX)	27.6	28.4	25.8	25.2	25.2	25.6
Detector	21.5	34.7	35.0	25.5	16.6	22.4
Reinteraction	4.2	8.6	4.1	5.1	3.7	3.6
MC stat	3.8	3.7	4.8	3.6	2.4	2.5
EXT stat	1.0	1.2	1.2	1.0	0.6	0.7
Data stat	13.0	13.0	16.7	12.3	8.4	9.0
POT + targets	3.8	4.0	3.6	3.5	3.5	3.5
Total (incl. data stat)	40.5	50.5	49.5	40.5	33.7	37.2

Table 17: Combined FHC+RHC systematic breakdown by source, bin-averaged fractional uncertainty (%); same provenance as Table 15.

Source	p_μ	p_π	$\cos \theta_\mu$	$\cos \theta_\pi$	$\theta_{\mu\pi}$	θ_μ
Prediction total	34.9	47.9	37.3	37.1	33.2	34.4
Cross section (GENIE)	10.3	15.7	10.4	10.6	10.0	9.3
Flux (PPFX)	24.2	30.2	25.7	25.6	25.7	25.2
Detector	21.4	30.8	23.8	22.5	17.7	20.6
Reinteraction	4.4	10.8	3.8	5.1	3.5	3.7
MC stat	3.9	3.7	2.9	2.9	1.9	2.4
EXT stat	0.8	1.6	1.2	1.0	0.6	1.0
Data stat	12.4	11.8	9.3	9.3	6.3	7.7
POT + targets	3.1	4.1	3.5	3.4	3.4	3.3
Total (incl. data stat)	37.2	49.5	38.6	38.3	33.8	35.4

6 Binning Optimisation, Systematic Breakdown, and Threshold Validation

This section records the binning optimisation on which the released binnings are based. Its table gives the binnings of the optimisation stage (p_π five bins, $\theta_{\mu\pi}$ seven, $\cos \theta_\pi$ five), which the response and A_C tests of §1.1 and §6.3 then coarsened or re-binned; the released binnings are those of the analysis note’s appendix.

6.1 Momentum-threshold validation

The two signal-definition momentum thresholds sit at the selection efficiency turn-on, measured against a threshold-free true signal (the CC ν_μ $1\mu 1\pi^\pm$ topology with the momentum/angle cuts removed):

Threshold	below	above	turn-on
$p_\mu > 0.15$ GeV/c	$\sim 0\text{--}1.5\%$	$\sim 9\text{--}12\%$	$0.145 \rightarrow 0.155$
$p_\pi > 0.175$ GeV/c	$\sim 0\text{--}3\%$	$\sim 12\text{--}16\%$	$0.165 \rightarrow 0.185$

Both are correctly placed, lowering them would add near-zero-efficiency phase space, raising them would discard measurable signal. The reconstructed pion momentum (muon-hypothesis range) is biased low and increasingly so with momentum (mean reco–true ≈ 0 at threshold, -0.15 GeV/c at 0.40) from the μ -vs- π mass hypothesis plus pion re-interaction; a pion-hypothesis mass correction is applied to the p_π reco bins.

6.2 Binning optimisation

Bins were designed to be at least the local RMS detector resolution wide (so the response stays well conditioned and the Wiener-SVD does not amplify the correlated flux/cross-section uncertainty) and to hold ≥ 45 selected-signal events (statistical floor). This lowers the total fractional uncertainty in every observable, chiefly through the statistical and unfolding-amplified terms; the flux/beamline term is correlated across bins and sets a floor:

Observable	bins	Total before	Total after
p_μ	$22 \rightarrow 7$	33.2%	25.0%
p_π	$5 \rightarrow 5$	27.7%	27.0%
$\cos \theta_\mu$	$12 \rightarrow 5$	31.3%	26.7%
$\cos \theta_\pi$	$12 \rightarrow 5$	48.5%	33.9%
$\theta_{\mu\pi}$	$9 \rightarrow 7$	28.4%	23.5%

6.3 Re-check of every released binning

The binnings of the optimisation stage were coarsened where A_C smearing left no resolved information ($\cos \theta_\pi$ $5 \rightarrow 4$, $\theta_{\mu\pi}$ $7 \rightarrow 5$). Because the normalisation, the weights, the beam-off

treatment, the fake-data throw and the detector variations that condition the response were all settled afterwards, every released binning was then re-tested, on the release preceding the $\cos\theta_\pi$ decision below, for whether finer or more bins are admissible. Two stages were used. First a pre-screen on the processed files (`macros/binning_screen.C`): for the released binning and finer candidates of each observable, the column-normalised migration diagonal per true bin, the efficiency per bin and the selected events per reconstructed bin, with the per-run scales and the central-value weight, and the 0.68 diagonal criterion applied to every bin. Second, the candidates that passed were built and unfolded in full (universes, covariance, closure; SLURM 3277770 and 3277778) and compared with the released extraction on the A_C row sums, the A_C conditioning ($s_{\min}/s_1$), the data-statistical term, the per-bin total uncertainties and the closure ratios.

Table 18: (Binning re-check on the release preceding the $\cos\theta_\pi$ decision: “released” means the binning of that time; the forward-split five-bin $\cos\theta_\pi$ row is the released binning.) Pre-screen: smallest column-normalised migration diagonal over the true bins of the released binning and of each finer candidate, FHC / RHC. The two-bin p_π row is evaluated here with a lower edge on the first reconstructed bin, which the released configuration does not have (the framework’s own diagonals are 0.91 and 0.76); the finer p_π candidates fail in their upper bins regardless.

Observable	Binning	min. diagonal FHC / RHC	criterion
p_μ	released 7	0.33 / 0.31	fails already (MCS resolution; §6)
	8, 9 bins	0.17 / 0.18, 0.22 / 0.22	fail
$\cos\theta_\mu$	released 5	0.68 / 0.68	at the limit
	6, 7 bins	0.64 / 0.63, 0.61 / 0.61	fail
θ_μ	released 5	0.67 / 0.65	at the limit
	7 bins	0.63 / 0.62	fail
$\cos\theta_\pi$	released 4	0.81 / 0.83	pass
	5 (backward split)	0.77 / 0.77	pass
	5 (forward split)	0.74 / 0.76	pass
	6 bins	0.73 / 0.75	pass
$\theta_{\mu\pi}$	released 5	0.79 / 0.79	pass
	6, 7 bins	0.74 / 0.73, 0.75 / 0.73	pass
p_π	released 2	0.58 / 0.56 (see caption)	—
	3, 4 bins	0.39 / 0.40, 0.29 / 0.29	fail

Table 19: (Binning re-check on the release preceding the $\cos\theta_\pi$ decision: “released” means the binning of that time; the forward-split five-bin $\cos\theta_\pi$ row is the released binning.) Full extraction of the candidates that passed the pre-screen, FHC / RHC: range of the A_C row sums, A_C conditioning $s_{\min}/s_1$, bin-averaged data-statistical term, range of the per-bin total uncertainty, and range of the per-bin closure ratio to the A_C -smeared truth. Backward split: edges $-1, -0.3, 0.2, 0.5, 0.75, 1$; forward split: $-1, -0.1, 0.35, 0.6, 0.8, 1$; six bins: $-1, -0.4, 0, 0.3, 0.55, 0.75, 1$; $\theta_{\mu\pi}$ six bins: $0, 0.55, 0.75, 0.95, 1.3, 1.85, 2.6$ rad; seven bins: $0, 0.5, 0.7, 0.9, 1.15, 1.45, 1.9, 2.6$ rad.

Binning	A_C row sums	$s_{\min}/s_1$	data stat	per-bin unc.	closure
$\cos\theta_\pi$ rel. 4	0.71–0.82 / 0.79–0.86	0.27 / 0.24	12.0 / 11.5%	30–49 / 32–55%	0.95–1.09 / 0.94–1.07
backward split 5	0.65–0.76 / 0.63–0.80	0.20 / 0.27	16.0 / 15.2%	29–73 / 34–91%	0.83–1.08 / 0.87–1.03
forward split 5	0.68–0.77 / 0.70–0.84	0.46 / 0.38	12.7 / 12.3%	30–53 / 33–48%	0.89–1.07 / 0.90–1.05
6 bins	0.73–0.84 / 0.77–0.89	0.18 / 0.15	19.0 / 16.3%	32–91 / 32–54%	0.66–1.13 / 0.81–1.23
$\theta_{\mu\pi}$ rel. 5	0.52–1.03 / 0.58–1.15	0.008 / 0.002	8.2 / 8.4%	35–40 / 30–39%	0.95–1.06 / 1.03–1.08
6 bins	0.56–1.29 / 0.58–1.36	0.002 / 0.000	8.4 / 10.2%	34–40 / 30–41%	0.95–1.06 / 1.03–1.11
7 bins	0.63–1.41 / 0.65–1.06	0.000 / 0.000	11.5 / 11.2%	37–40 / 32–46%	0.85–1.04 / 0.87–1.14

Conclusions. p_μ , $\cos\theta_\mu$, θ_μ and p_π cannot go finer: their released binnings already sit at or below the diagonal criterion, set by the MCS momentum resolution and the angular resolution against the neutrino direction, and every finer candidate fails it. $\theta_{\mu\pi}$ admits six or seven bins by the diagonal and closes acceptably, but the additional bins carry no resolved information: the

smallest singular value of A_C , already 0.008 of the largest at five bins, goes to zero, and the row sums overshoot to 1.3–1.4 in the bins the regularisation has to fill in, so the released five bins stand. $\cos\theta_\pi$ is the one observable with room, and only in the forward hemisphere where the events are: splitting the backward region produces a bin of 73–91% uncertainty that closes at 0.83–0.87, and six bins do the same, whereas the forward split (edges $-1, -0.1, 0.35, 0.6, 0.8, 1$) keeps every diagonal above 0.74, keeps the per-bin uncertainties in the released range, raises the data-statistical term by only 0.7% and *improves* the conditioning of A_C ($s_{\min}/s_1$ from 0.27 to 0.46 in FHC, 0.24 to 0.38 in RHC) while closing to 0.89–1.07. It is the only change this re-check supports, and it is the released binning: the generator predictions are computed at these edges from the event-level files (the unchanged observables reproduce the previous prediction files to machine precision, which also verifies the generator chain end to end), and the D’Agostini cross-check is run on it. No control-region distribution uses $\cos\theta_\pi$, so the frozen sideband binning is unaffected. The proton-tagged $\cos\theta_\pi$ keeps its four bins. The released binnings of the proton-tagged observables were re-tested in 2026-09, on the maximin scan and against a relaxed 0.50 diagonal criterion as well as the nominal 0.68: twenty-six candidate binnings pass the relaxed diagonal and every one of them fails on the conditioning of A_C . The single exception is a two-bin W_{had} (0.58/0.54), which is better conditioned than the released six-bin scheme and was nevertheless not adopted, since two bins buy that conditioning by discarding the shape W_{had} is measured for. No released proton-tagged binning changes. The one binning that did change is the new θ_p , whose edges were taken from the RHC chain rather than the FHC one after the scan was run on both: at the same four bins they condition three to four times better in both usable configurations.

7 Consistency between configurations and with the sibling ν_e measurement

7.1 Consistency with the sibling ν_e CC 1π measurement

This analysis is one of a pair of MicroBooNE NuMI CC single-charged-pion measurements; the companion ν_e CC 1π analysis shares the same cross-section-extraction framework, beam and reconstruction. A cross-check against its internal note confirms consistency on the essential conventions and identifies the differences that a combined NuMI CC π programme should reconcile.

Consistent by construction. Both use the Wiener–SVD extraction framework; the updated NuMI flux with the fixed geometry bug and Geant v4.10.4; the same signal topology (exactly one $\pi^\pm$, zero π^0 or heavier mesons, any number of protons, charged current), and the same systematic suite, PPFx flux hadronic universes (the 12 beamline-geometry unisims the ν_e analysis carries are evaluated but not in this analysis’s released covariance, Sec. 15 of the analysis note), the 600-universe GENIE cross-section multisim with unisims, the Geant4 reinteraction reweighting, the standard detector-variation set, and 2% / 1% / 100% POT / target / dirt normalisation uncertainties. The FHC beam-on data POT per run agrees exactly (Run 2 1.268, Run 4 2.075×10^{20}).

Differences reconciled or flagged. (i) **POT accounting:** the good-POT is the same beam data for the ν_e and ν_μ analyses, so the exposures are identical; Run 1 FHC = 3.283×10^{20} (standard trigger; the open-trigger era is treated separately), giving a total FHC exposure of 8.857×10^{20} over Runs 1–5. (ii) **Run 4:** the Run4 overlay is identical to Run4c+Run4d combined (same 859,368 events and 2.834×10^{21} POT); only one may be used to avoid double counting. (iii) **Pion threshold:** ν_e uses $KE_\pi > 40$ MeV ($p_\pi > 0.113$ GeV/c) versus our $p_\pi > 0.175$ GeV/c (Table 8 of the analysis note). (iv) **Opening-angle cut:** ν_e excludes $\theta_{e\pi} > 170^\circ$ (a shower-specific back-to-back misreconstruction), whereas our muon-track topology uses $\theta_{\mu\pi} < 2.6$ rad (149°). (v) **Regularisation:** the ν_e analysis smooths on the first derivative, this analysis on the second (§5.1). (vi) **Scope:** both analyses cover Runs 1–5 in FHC and RHC; the present measurement

reports FHC, RHC, and their combination.

7.2 Is the combined configuration the combination of its own inputs?

The integrated cross section extracted from a given configuration is not a configuration-independent number: it depends on the additional smearing matrix A_C , which differs per configuration and per observable, so comparisons across configurations are made observable by observable and against the fake-data truth rather than between measurements. Two tests establish that the combined configuration is what it should be.

The combined truth against the exposure-weighted average of its inputs. A configuration spanning both horn currents should predict a cross section between those of its two inputs. Comparing the combined fake-data truth with the POT-weighted mean of the FHC and RHC truths, observable by observable, gives a mean ratio of 1.027 with a range of 0.958–1.071 for the inclusive family and 1.020 (0.989–1.052) for the proton-tagged family (measured on one fake-data pass; the ratio is a property of the truth and the normalisation, not of the throw). The test is sensitive: it is what requires the two per-mode detector-variation sets of the combined configuration to be pooled as one sample scaled once to the data exposure, since scaling each set to the full exposure and summing would double the detector covariance, over-regularise the combined extraction and push its truth ratio to 0.91 with a worst observable at 0.77. With the pooled treatment the combined detector term lies between or close to those of its inputs (analysis note, Tables 17–19: 21.4% against 24.7% and 21.5% for p_μ).

The RHC deficit is in the input, not the extraction. The one-bin totals of the release give an RHC/FHC ratio of the injected tune of 0.901 (inclusive) and 0.878 (proton-tagged), and the extracted one-bin ratios are 0.908 and 0.860 (Table 21 of the analysis note). Normalisation is not implicated: the exposures sum exactly ($8.857 \times 10^{20} + 1.1082 \times 10^{21} = 1.9939 \times 10^{21}$ POT) and the per-configuration flux values reproduce the POT-weighted combination to 0.01%. The deficit is therefore a property of the $\bar{\nu}_\mu$ -dominant sample, and the number to be compared with data at unblinding is the tune ratio.

External reproduction of the released curves. The transformation $(d\sigma/dx)_i = w_i^{-1} \sum_j (A_C)_{ij} p_j$ applied to the released truth-level predictions reproduces every released generator curve to a worst per-bin deviation of 1.2×10^{-10} . This is part of the release procedure because it is the one check that an internal closure cannot make: the unfolded data and the truth can both be right while the model comparison is wrong, and only an external user’s arithmetic exposes it.

8 Material supporting the analysis note’s selection and validation sections

The studies in this section are referred to from the analysis note; they are kept here so that the note carries only the current-release argument.

8.1 Selection-optimisation study

Why the phase space and the opening-angle cut are what they are. The measured phase space and the $\theta_{\mu\pi} < 2.6$ rad cut were chosen on the Run-1 FHC sample with a reference phase space of $p_\mu, p_\pi > 0.1$ GeV/c; the numbers in this paragraph and in Table 20 are that study and document the choice. The full-exposure yields, efficiency and purity of the released measurement are those of Table 11 of the analysis note above and are quoted from there.

- Signal MC (POT-weighted): 324.7 events
- Total background: 243.6 (≈ 199.7 ν -bkg + 40.1 EXT + 3.8 dirt)
- True signal generated (efficiency denominator): 2577–2684 events (observable-dependent)
- **Signal purity: 57.1%**
- **Signal efficiency: 12.1–12.6%**
- **Measured phase space (reported result): efficiency 15.6%, purity 54.9%** (reference phase-space study: selected signal 311.9, generated 1999, background 256.4; Table 20)

Per-run scaling. The yields above and in Table 11 of the analysis note use the per-run POT factors of Table 5 of the analysis note (each run scaled by its own $D_{\text{run}}/\text{MC}_{\text{run}}$), and the EXT by the per-run trigger ratios (per-run gate ratios, Sec. 5 of the analysis note). A single per-mode factor would differ from the per-run scheme by about 2% in the FHC final-cut signal, a small and nearly uniform re-weighting of the run mixture that moves efficiency and purity by $< 0.5\%$; the per-run scheme is used because it is what real beam-on data require. The fake data are thrown from the same per-run-scaled prediction, so the closure tests the bookkeeping of the scheme, not its correctness against data.

Measured phase space and efficiency turn-on. The reference selection above has near-dead efficiency at low momentum and at the large opening-angle extreme: from the fine truth distributions, the per-bin efficiency is only $\sim 2\text{--}5\%$ for $p_\mu, p_\pi < 0.15$ GeV/ c (the 10/20 cm track-length cuts) and the region above the reco $\theta_{\mu\pi} = 2.6$ rad cut is dominated by cosmics ($\sim 7\%$ purity). The measured phase space $p_\mu > 0.15$, $p_\pi > 0.175$ GeV/ c , $\theta_{\mu\pi} < 2.6$ rad (Sec. 6 of the analysis note) raises and flattens the efficiency to $\sim 13\text{--}20\%$ (Fig. 18 of the analysis note). Table 20 compares the selection across the iterations. Two conclusions: (i) the 2.6 rad *upper* cut is kept: on the full exposure the events it removes are exactly those of the cosmic control region (Table 23 of the analysis note), in FHC 6.0 signal, 69 neutrino background, 53 beam-off and 2.2 dirt, a signal purity of 4.6%; dropping it would lower the signal-region purity from 58.5% to 54.5%, raise its beam-off content from 30 to 83 events and recover only 2.7% of the generated signal ($2055 \rightarrow 1999$ in the table), and the region it removes is more useful as the beam-off control region; (ii) a *lower* opening-angle cut is counterproductive, the small-angle region is high-purity ($\sim 58\%$), so cutting it lowers the selected signal without improving purity, which is why no lower cut is used.

Table 20: Selection-optimisation study across the phase-space iterations, on the Run-1 FHC sample. Every entry is Monte-Carlo derived (GENIE overlay signal and background, EXT, dirt); the thresholds chosen here are those of the full-exposure selection of Table 11 of the analysis note. The beam-off column uses the pooled beam-off scaling of that study, so its absolute values are not those of the full exposure; the ranking of the options is unaffected. “no upper cut” drops the $\theta_{\mu\pi} < 2.6$ cut; “[0.314, 2.6]” adds a lower OA cut. The final choice keeps the 2.6 rad upper cut with no lower cut.

	Reference ($p > 0.1$)	no upper cut ($p > .15/.175$)	[0.314, 2.6] (lower cut)	Final ($\theta < 2.6$)
Generated signal	2684	2055	1916	1999
Selected signal	324.7	320.3	301.8	311.9
Efficiency	12.1%	15.6%	15.8%	15.6%
EXT (cosmic)	40.1	136.2	38.0	40.1
Total bkg	243.6	385.4	252.3	256.4
Purity	57.1%	45.4%	54.5%	54.9%
sig $\times$ purity	185.6	145.4	164.4	171.2

8.2 Cosmic-direction cuts considered and rejected

A directional cut against residual cosmics: considered and rejected. Part of the surviving beam-off sample is thought to be broken cosmic muon tracks, where the break is reconstructed as a vertex and the track direction bears no relation to it. That leaves a directional signature, and it is present in the data: among selected events the beam-off sample is enriched at backward reconstructed muon angles by nearly an order of magnitude relative to signal.

reco $\cos \theta_\mu$	EXT (beam-off)	MC signal	enrichment
< 0	31.8%	4.1%	$7.8\times$
< -0.5	9.1%	1.4%	$6.4\times$

The enrichment is real but the arithmetic does not support a cut. At the final cut the FHC sample is 943 signal, 30 EXT and 1612 predicted in total (Table 11 of the analysis note). A $\cos \theta_\mu < 0$ requirement would remove $\approx 32\%$ of the EXT, about 10 events, and $\approx 4\%$ of the signal, about 39 events. Because the beam-off sample is subtracted from the data directly, and that subtraction is unbiased by construction, removing part of it buys only a reduction in the subtraction’s *statistical* uncertainty, which is dominated by the 8 raw Run-1 events scaled by 2.1 (≈ 6 events for the 30), to ≈ 5 events. Spending 39 signal events to gain ≈ 1 event of uncertainty is a net loss.

The same bound applies to any cosmic cut, not just this one: EXT is 1.9% of the selected sample and is already subtracted without bias, so removing *all* of it for *zero* signal loss would gain about 6 events of statistical uncertainty on a sample of 1612. Any cut costing more than about 1% of the signal is a net loss regardless of how efficiently it tags cosmics. The residual beam-off contribution is therefore left in place and handled by subtraction.

A more surgical variant remains untested. The CC0 π analysis tags mis-directed tracks with a calorimetric direction-preference flag in coincidence with a poor forward-muon fit, $\chi_\mu^2 > 6$, which fires on tracks whose dE/dx profile disagrees with the assigned direction rather than on backward-going tracks as such. That would not cut genuine backward muons, which outnumber mis-reconstructed ones here by roughly 8:1. Both quantities exist in the input ntuples but are not carried into the processed files. It is bounded by the same ≈ 6 -event ceiling on the EXT statistical uncertainty, so it is worth measuring but cannot become a significant improvement.

8.3 Combined-configuration reconstruction-level spectra

The selected and background-subtracted spectra of the combined configuration, companions to the FHC and RHC ones of the analysis note.

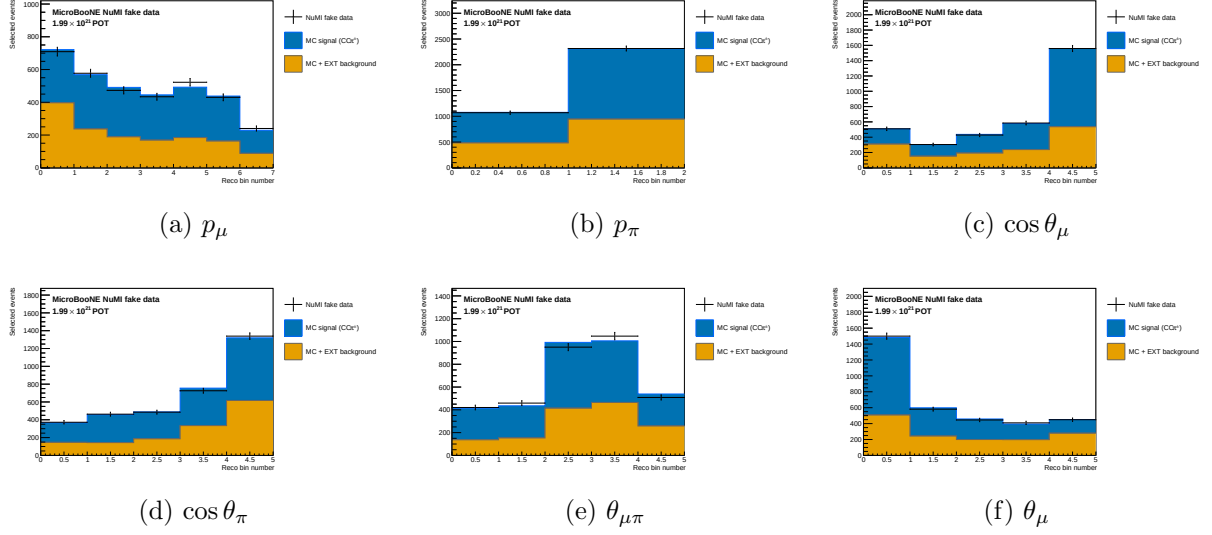


Figure 6: Stage 1, combined: selected reconstructed spectra at the combined FHC+RHC exposure.

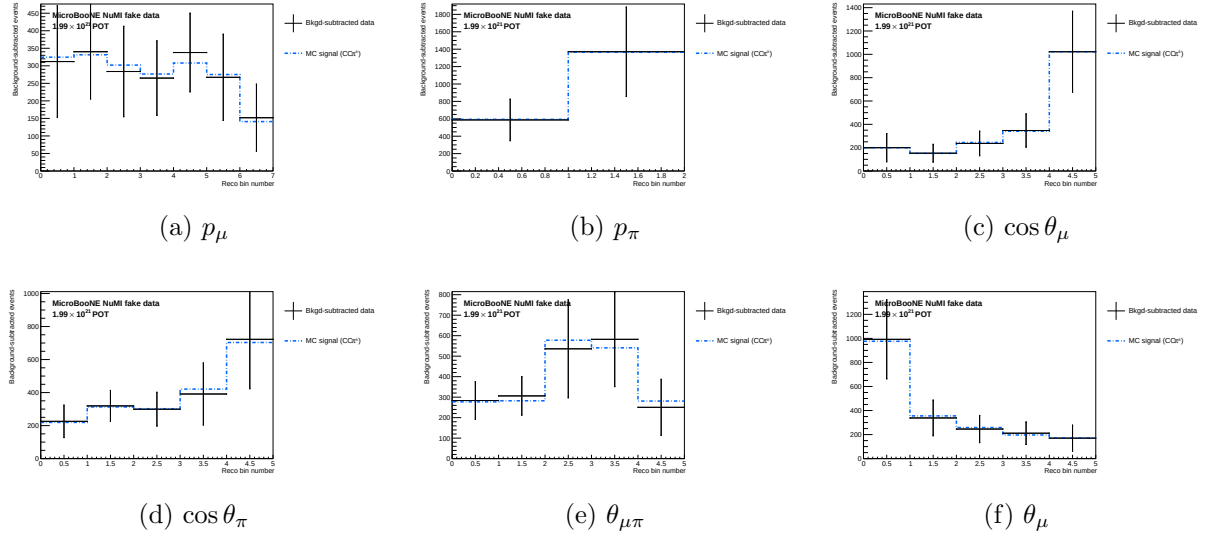


Figure 7: Stage 2, combined: background-subtracted data compared with the central-value MC signal.

8.4 The rejected five-bin p_π response

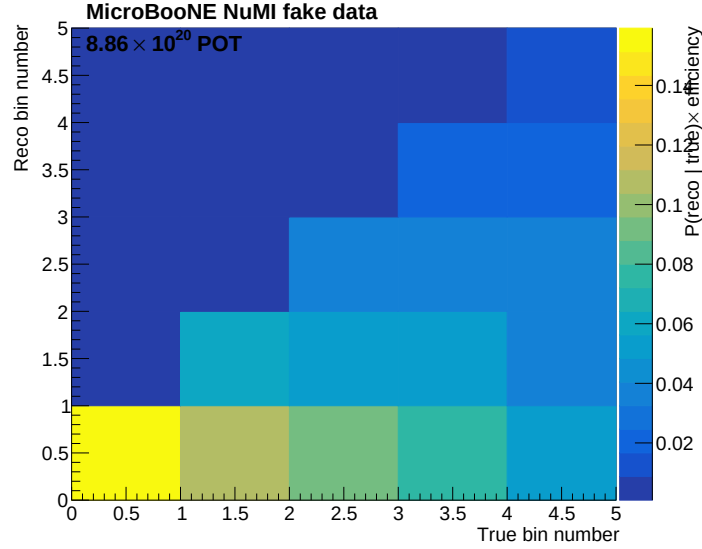


Figure 8: Smearceptance matrix of the rejected five-bin p_π scheme (FHC): reconstructed events pile into the lowest bin from every true bin, the diagnostic that motivated the two-bin scheme of §1.1; the released two-bin response is in the analysis note.

8.5 A_C row sums of every released extraction

Row sums of the released additional-smearing matrices, inclusive and proton-tagged, and the two matrices with a negative row.

Observable	FHC	RHC	combined
<i>Inclusive</i>			
p_μ	0.66–0.91	0.58–0.99	0.68–1.01
p_π (2-bin)	0.35–0.54	0.40–0.51	0.29–0.54
$\cos \theta_\mu$	0.17–0.39	0.05–0.45	0.09–0.43
$\cos \theta_\pi$	0.68–0.77	0.70–0.84	0.71–0.83
$\theta_{\mu\pi}$	0.52–1.03	0.58–1.15	0.63–1.21
θ_μ	0.58–0.78	0.22–1.16	0.53–0.67
<i>Proton-tagged</i>			
p_μ	0.46–1.00	0.55–0.89	0.65–0.96
p_π (2-bin)	0.34–0.84	0.41–0.45	0.47–0.48
$\cos \theta_\mu$	0.61–0.70	−0.08–0.57	0.28–0.46
$\cos \theta_\pi$	0.66–0.77	0.62–0.77	0.65–0.76
$\theta_{\mu\pi}$	0.37–0.87	0.31–0.75	0.35–0.78
$W_{\pi p}$	0.45–1.22	−0.14–1.26	0.32–1.24
W_{had}	0.54–0.81	0.30–0.86	0.18–0.87
δp_T	0.54–0.86	0.45–0.82	0.47–0.87
$\delta \phi_T$	0.65–0.78	0.53–0.72	0.28–0.67
$\delta \alpha_T$	0.59–0.98	0.57–0.93	0.60–0.96
p_n	0.71–0.86	0.61–0.82	0.59–0.86

Three features matter for interpretation.

The suppression is large: inclusive $\cos \theta_\mu$ has a row sum as low as 0.05 in RHC, a strong suppression of that output mode for a flat input. It also differs strongly between observables,

which is precisely why the integrated cross section obtained by summing a differential result varies from observable to observable while the underlying physical total cannot. The D’Agostini cross-check, whose integral is flat across observables, is what establishes that this spread is the smearing rather than a physical or acceptance effect.

Two extractions have *negative* row sums, and the pattern identifies their cause. In $W_{\pi p}$ the top row carries negative off-diagonal terms in *all three* configurations (summing to -0.14 FHC, -0.30 combined, -0.61 RHC against diagonals of 0.73 , 0.62 and 0.47); only RHC tips negative overall (row sum -0.14), because its diagonal element is the smallest of the three. The same happens in the first $\cos\theta_\mu$ bin of the proton-tagged RHC sample, whose diagonal is 0.079 against off-diagonal terms summing to -0.16 (row sum -0.08).

So this is not an RHC-specific defect but the regularisation pushing back against the least-determined bin of each observable — for $W_{\pi p}$, the same top bin that is only 2% diagonal in the six-bin response and that already led to withdrawing the differential shape. A negative row is not forbidden, since A_C is a regularisation operator rather than a probability matrix, but a prediction smeared through one is being told that raising the truth in that bin *lowers* the measured content, which is a statement about the regularisation and not about the detector. Treat those two rows with corresponding caution.

8.6 Ensemble study: reference construction and full-covariance pulls

Ensemble coverage: are the uncertainties the right size? The closure test above uses a single throw. It can show that the extraction reproduces its input, but a single residual has no width, so it cannot say whether the quoted uncertainty is correctly sized. To answer that, thirty-two statistically independent Poisson realisations of the FHC data were each put through the *whole* chain — thrown, universes rebuilt from that throw, unfolded — and the per-bin pull

$$p_i = \frac{\hat{x}_i - (A_C x_{CV})_i}{\sigma_i}$$

formed against the fixed central-value truth smeared by that member’s own A_C . (The per-throw *fake-data* truth is not a valid reference: it follows the Poisson draw, so numerator and reference fluctuate together and the width comes out too small.)

Three extractions were run on the current release, each with the same 32 throws (with the weight rule of Eq. 4 of the analysis note), chosen to bracket the conditioning: the well-behaved p_μ ; $\cos\theta_\mu$, whose A_C row sums fall as low as 0.17 in FHC (0.050 in RHC) so that the regularisation removes up to 83% (95%) of a bin’s normalisation; and the one-bin total, which has no regularisation at all. If the uncertainties were mis-sized anywhere, the second is where it would show; if the extraction carried a bias independent of the regularisation, the third is where it would show.

Reference covariance	pull mean	pull width	68% / 95% coverage
<i>statistical only</i>			
$d\sigma/dp_\mu$	-0.05	1.08	$62.1\% / 92.0\%$
$d\sigma/d\cos\theta_\mu$	-0.18	0.97	$71.2\% / 93.1\%$
one-bin total	-0.25	1.02	$62.5\% / 93.8\%$
<i>full</i>			
$d\sigma/dp_\mu$	-0.01	0.35	$99.1\% / 100\%$
$d\sigma/d\cos\theta_\mu$	-0.06	0.33	$99.4\% / 100\%$
one-bin total	-0.03	0.11	$100\% / 100\%$
expected	0	1	$68\% / 95\%$

Against the statistical covariance all three widths are consistent with unity — 1.08 , 0.97 and 1.02 , each determined to $\pm 13\%$ ($1/\sqrt{2 \times 32}$) with this many members — and the interval coverage

is correct within the same statistics. **The statistical uncertainties are correctly sized, and the strongly suppressed observable behaves no differently from the well-conditioned one.** That the severe A_C suppression in $\cos\theta_\mu$ does not distort its uncertainties is the more informative of the two results, since p_μ passing was the expected outcome.

Two further things the ensemble shows. Against the *full* covariance the pull width is 0.33–0.35 for the differential results and 0.11 for the total, and coverage is essentially 100%: the quoted uncertainty is about three times the throw-to-throw spread, because the ensemble varies only the data statistics while the denominator carries flux, detector and cross-section terms that are identical in every member. That is the same effect as the closure p -values clustered near unity, now measured rather than argued.

Second, the ensemble means sit below the reference in all three cases: by 0.5% for p_μ (0.8074 against 0.8113, 0.4σ on the mean), 1.0% for the one-bin total (1.0206 against 1.0309, 1.3σ) and 2.7% for $\cos\theta_\mu$ (0.7704 against 0.7918, 1.9σ), with per-bin pull means of -0.05 , -0.25 and -0.18 (the first $\cos\theta_\mu$ bin at -0.44). The three share the same 32 throws and are therefore correlated, so they are one measurement of a small negative offset, not three. The throw itself is unbiased: the mean selected count over the 32 members is 1578.7 against the neutrino-MC expectation of 1580.6 (ratio 0.999), so whatever offset remains is in the extraction, not in the pseudo-data. The offset is at most two standard deviations of the ensemble mean and is largest where the regularisation is strongest, but the one-bin total, which has no regularisation, also sits low, so the excess in $\cos\theta_\mu$ over the common $\approx 1\%$ is the only part that can be attributed to A_C . No correction is applied and no uncertainty is assigned; a larger ensemble would settle whether the common offset is real.

The study covers three extractions in one configuration (FHC). Extending it to RHC and to the proton-tagged family would test whether the same holds where the statistics are sparser; the machinery is in place.

A Identity closure: unfolding the prediction with itself

The fake-data closure tests reported in the body compare an unfolded Poisson throw against the truth it was thrown from. They therefore test the whole chain at once, and a modest χ^2 can hide a genuine inconsistency inside the unfolding itself, because the statistical fluctuation of the throw is the dominant term.

The test here removes that fluctuation. Wiener-SVD returns $\hat{x} = Ur$ for a measured reco vector r , and defines the additional smearing matrix as $A_C = UR$, where R is the response (smearacceptance) matrix. Feeding an *exact* prediction $r = Rt$ back through the chain must therefore return

$$\hat{x} = URt = A_C t \tag{1}$$

identically, for *any* true vector t . Rather than pick one t , we test the matrix identity directly:

$$\|UR - A_C\|_\infty \stackrel{!}{=} 0. \tag{2}$$

This is stronger than a single-vector test and independent of the data. It is sensitive to the failure mode a fake-data test cannot see: the unfolding matrix and the smearing matrix being built from different responses, or from responses evaluated in different universes, which would bias every extracted cross section by a fixed factor while leaving the closure χ^2 untouched.

The check is implemented inside `UnfolderNuMI` and runs on every extraction, printing a `[IDENTITY]` line. The release has 51 extractions (six inclusive observables and eleven proton-tagged ones, each in three beam configurations) and every one of them passes the same check. The worst deviation over the 51 extractions of the release is 4.4×10^{-6} relative, on FHC $W_{\pi p}$; the inclusive observables are all below 2.5×10^{-7} . These are the rounding errors of the singular-value decomposition and the matrix products that follow it, not physical effects: they remain five orders of magnitude below the smallest systematic uncertainty in the analysis.

The values are larger than the 10^{-10} level seen before the regularisation matrices were made width-aware, and the reason is arithmetic rather than physical. The width-aware operator carries factors of $1/w_j$ that span decades across a binning with a 14:1 width ratio, so the matrix products accumulate more floating-point error. $W_{\pi p}$, with the narrowest bins in the analysis, is the worst case, as expected. Nothing about the identity is weaker; only the arithmetic that verifies it is noisier. The unfolding construction is therefore exactly self-consistent, and every deviation seen in the fake-data closure tests is attributable to the statistical throw and to the response model, not to the unfolding implementation.

B Physics background and prior measurements (extended)

B.1 Production Mechanisms

At neutrino energies of 0.5–5 GeV, charged-current single charged-pion production proceeds through three mechanisms.

Baryon resonance production dominates at $E_\nu \lesssim 2$ GeV. The neutrino excites the struck nucleon to a baryon resonance, most prominently the $\Delta(1232)$ isobar, which then decays to a nucleon and pion:

$$\nu_\mu n \rightarrow \mu^- \Delta^{++} \rightarrow \mu^- \pi^+ p, \quad \nu_\mu p \rightarrow \mu^- \Delta^+ \rightarrow \mu^- \pi^0 p \text{ (or } \pi^+ n). \quad (3)$$

The standard model is due to Rein and Sehgal [10], which uses the Feynman–Kislinger–Ravndal (FKR) quark model to compute form factors for all baryon resonances up to $W \lesssim 2$ GeV. The modern implementation used in GENIE for resonance (RES) production is the Kuzmin–Lyubushkin–Naumov Berger–Sehgal (KLN-BS) model [11, 12, 13, 14], which combines the lepton-mass-corrected vector and axial form factors of Kuzmin *et al.* [11] with the updated $\Delta(1232)$ electromagnetic form factors of Graczyk and Sobczyk [12] and the improved treatment of the Δ line shape and the lepton-mass effects of Berger and Sehgal [13]. (Berger and Sehgal also discuss the interference between the resonant and non-resonant amplitudes; that interference term is not implemented in the GENIE v3 configuration underlying the MicroBooNE tune [2], where the non-resonant contribution is added incoherently.) Higher resonances ($N(1440)P_{11}$, $N(1520)D_{13}$, etc.) are included up to $W \lesssim 2$ GeV and contribute at $E_\nu \gtrsim 1.5$ GeV.

Non-resonant and DIS backgrounds. A non-resonant continuum contributes at all W ; in a full amplitude-level treatment it interferes (constructively or destructively by isospin) with the resonant amplitude, although the generators used here add it incoherently. Deep-inelastic scattering (DIS) becomes significant above $W \approx 1.6$ GeV. Together these processes represent $\sim 20\%$ of the $\text{CC}\pi^\pm$ rate in the energy range of this analysis and are modelled in GENIE via the Bodek–Yang formalism.

Coherent pion production. The neutrino can scatter coherently off the entire nucleus [7, 8], $\nu_\mu A \rightarrow \mu^- \pi^+ A^*$, leaving the nucleus in its ground state. The cross section is small ($\lesssim 10\%$ of the total) below 5 GeV. The signal definition of Sec. 6 of the analysis note is a final-state topology (one muon, one charged pion, no other mesons) and carries no truth-level veto on the interaction mode, so coherent $\text{CC}\pi^\pm$ events that satisfy it are *included in the signal*; they are a small signal component, not a background.

B.2 Nuclear Effects and Final-State Interactions

On a heavy nucleus such as argon ($A = 40$), two classes of nuclear correction modify the $\text{CC}\pi^\pm$ rate and kinematics significantly.

Initial-state effects. Fermi motion of the struck nucleon broadens kinematic distributions, and Pauli blocking suppresses transitions to low-momentum nucleon final states near threshold. Nuclear binding also shifts the effective nucleon mass relative to free-nucleon predictions.

Final-state interactions (FSI). The produced pion must traverse the nuclear medium before being detected. Intranuclear rescattering occurs via:

- *Pion absorption* ($\pi A \rightarrow NN \dots$): probability $\sim 25\text{--}30\%$ on Ar; converts $\text{CC}\pi^\pm$ events into $\text{CC}0\pi$ topology.
- *Charge exchange* ($\pi^+ n \rightarrow \pi^0 p$): produces a $\text{CC}\pi^0$ event.
- *Quasi-elastic scattering*: pion survives but changes direction; modifies p_π and θ_π distributions.
- *Inelastic production* ($\pi N \rightarrow \pi\pi N$): rare below $p_\pi \approx 400 \text{ MeV}/c$.

FSI are modelled in GENIE via the INTRANUKE (hA) cascade. Because absorption and charge exchange mix $\text{CC}\pi^\pm$, $\text{CC}\pi^0$, and $\text{CC}0\pi$ topologies at the $\sim 30\%$ level, the signal in any $\text{CC}\pi$ analysis is necessarily defined by the final-state particle content observed in the detector, not by the underlying production process (Sec. 6 of the analysis note).

B.3 Generator Predictions and Known Tensions

Major neutrino event generators reproduce the inclusive CC cross section to 10–15% but show systematic spread in the $\text{CC}\pi^\pm$ rate, owing to differences in the resonance model, the treatment of the transition region between resonance and DIS, and the FSI cascade.

GENIE v3 [1, 3]. For resonance production GENIE implements the KLN-BS model [11, 12, 13, 14] for the full tower of 18 baryon resonances up to $W \lesssim 2 \text{ GeV}$, including the $\Delta(1232)P_{33}$, $N(1440)P_{11}$ (Roper), $N(1520)D_{13}$, $N(1535)S_{11}$, $\Delta(1620)S_{31}$, $\Delta(1700)D_{33}$, $N(1680)F_{15}$, $\Delta(1950)F_{37}$, and their isospin partners. The hadronic invariant mass range $1.0 < W < 1.7 \text{ GeV}$ is bridged by a resonance–DIS transition using the AGKY (Andreopoulos–Gallagher–Kehayias–Yang) model, which applies KNO scaling in the transition region and the Bodek–Yang DIS model above $W \approx 1.7 \text{ GeV}$. Non-resonant single-pion production is included through the low- W extension of the Bodek–Yang DIS model and is added incoherently to the resonant contribution; no resonant/non-resonant interference is implemented. Coherent π^+ production uses the Rein–Sehgal coherent model. FSI are handled by the INTRANUKE hA empirical model, in which each hadron undergoes at most one effective interaction whose probability and outcome are tuned to hadron–nucleus data; it is the alternative hN microscopic cascade, also available in GENIE, that propagates hadrons step-by-step using free-nucleon πN and NN cross sections. GENIE has been reported to over-predict the $\text{CC}\pi^\pm$ rate by 20–35% relative to bubble-chamber and MiniBooNE data in several comparisons when normalised to the BNL dataset; the tension is reduced but not eliminated in the AR23 tune.

NuWro [4]. NuWro models resonance production via the $\Delta(1232)$ only; no higher baryon resonances are included. The Δ is treated with the Paschos–Yu formalism for the Δ excitation amplitude, retaining the full angular structure of the pion decay. A smooth analytic interpolation connects the single- Δ resonance region to the DIS regime at higher hadronic invariant mass ($W \gtrsim 1.4 \text{ GeV}$), using Bodek–Yang structure functions for the deep inelastic contribution; the transition avoids a hard boundary between resonance and continuum. Non-resonant single-pion production is included as a separate amplitude. FSI are implemented using the microscopic Salcedo–Oset model, which treats pion propagation in nuclear matter through the pion self-energy in the random-phase approximation, naturally accounting for medium modifications such as Δ broadening and pion absorption via two-body processes ($\pi NN \rightarrow NN$) that are absent in purely cascade-based approaches. NuWro has historically shown better agreement with MiniBooNE and bubble-chamber data than GENIE, partly attributable to the absence of overestimated higher-resonance contributions.

NEUT [15, 6]. NEUT, the generator used by T2K and Super-Kamiokande, implements resonance production via a modified Rein–Sehgal model including 18 resonances up to $W \lesssim 2$ GeV, with Graczyk–Sobczyk [12] electromagnetic form factors for the $\Delta(1232)$. The axial mass M_A^{RES} is a tunable parameter; T2K applies a data-driven tune that reduces the overall $\text{CC}\pi^\pm$ normalisation by $\sim 10\text{--}20\%$ relative to the default. The transition to DIS is handled similarly to GENIE, using GRV98 LO parton distribution functions with Bodek–Yang corrections above $W \approx 2$ GeV. Coherent pion production follows the Rein–Sehgal coherent model with a Berger–Sehgal-style correction for finite lepton mass. FSI are modelled by a custom NEUT intranuclear cascade that tracks each produced hadron through the nucleus using step-by-step interactions sampled from πN cross sections, with nuclear density and mean-free-path corrections derived from the optical potential.

GiBUU [5]. The Giessen Boltzmann–Uehling–Uhlenbeck (GiBUU) generator takes a fundamentally different approach: instead of a probabilistic step-by-step cascade, it solves a coupled set of kinetic transport equations for the full hadronic system inside the nuclear medium. Resonance production includes the $\Delta(1232)$ and higher resonances whose in-medium spectral functions are broadened by collisional damping; the Δ width in the nuclear medium is substantially larger than its free-space value of $\Gamma_0 \approx 118$ MeV, suppressing threshold production. Pion propagation after production is governed by a mean-field potential (the pion optical potential), which shifts pion momenta as they traverse the nucleus – an effect absent in cascade models and particularly important for low-momentum pions ($p_\pi \lesssim 200$ MeV/ c) near the pion production threshold. The transition to DIS uses PYTHIA for string fragmentation above $W \approx 1.6$ GeV. GiBUU predictions for pion angular distributions and for the backward-hemisphere $d\sigma/d\cos\theta_\pi$ can differ substantially from cascade-based generators, making it a valuable cross-check of FSI systematics.

Because the present results are extracted on central-value fake data, no data/MC deficit is measured here. What the analysis does provide is the spread of the standalone generators in the measured phase space, a factor of ~ 1.6 between GENIE and NEUT (Table 14), which brackets the injected MicroBooNE tune; the over-prediction observed for several generator configurations in parts of the relevant phase space in the measurements surveyed below is the context for that spread, not a result of this note.

B.4 Previous Experimental Measurements

Table 21 summarises published $\text{CC}\pi^+$ measurements on nuclear targets ordered chronologically.

Table 21: Selected $\text{CC}\pi^+$ cross-section measurements on nuclear targets. $\langle E_\nu \rangle$ is the approximate mean or peak neutrino energy. “Type” indicates whether a total integrated (T) or differential (D) cross section was reported. “Target” gives the nucleus primarily used for cross-section extraction.

Experiment	Target	$\langle E_\nu \rangle$ (GeV)	Type	Key result / reference
ANL 12-ft [16]	D	1.0	T	Bubble chamber; foundational σ vs E_ν
BNL 7-ft [17]	D	1.5	T	Higher statistics; 15% normalisation tension with ANL
K2K [18]	C	1.3	T	First accelerator $\text{CC}\pi^+$ on carbon near detector
MiniBooNE [19]	CH_2	0.7	D	Q^2 , W , p_μ , θ_μ ; 20% excess over NUANCE
MINERvA [20]	CH	3.6	D	p_π , θ_π , Adler angles; NuMI medium-energy
T2K ND280 [23]	CH	0.85	D	p_μ , $\cos\theta_\mu$; off-axis beam similar to this work
ArgoNEUT [24]	Ar	9.6	T	First $\text{CC}\pi^+$ on Ar; 115 ν + 158 $\bar{\nu}$ events (1.25×10^{20} POT)
MicroBooNE BNB [26]	Ar	0.7	D	Full BNB dataset (1.11×10^{21} POT); $\sigma =$ (3.75 ± 0.80) $\times 10^{-38}$ cm ² /Ar
MicroBooNE NuMI ν_e [25]	Ar	0.73	D	First $\nu_e + \bar{\nu}_e$ $\text{CC}\pi^+$ on Ar; $\sigma = (0.93 \pm 0.28) \times$ 10^{-39} cm ² /nucleon

Deuterium bubble chambers: ANL and BNL (1979–1986)

The ANL 12-foot bubble chamber [16] and BNL 7-foot bubble chamber [17] provided the first high-statistics measurements of $\nu_\mu d \rightarrow \mu^- \pi^+ pp$ in the 1–3 GeV range. These measurements are important because the deuterium target is close to a free nucleon, minimising nuclear corrections relative to heavier targets. The ANL and BNL total cross sections disagree by $\sim 15\%$ in absolute normalisation, a discrepancy long attributed to flux uncertainties; the BNL flux is now believed to be better constrained. This ANL–BNL tension directly propagated into the spread of generator predictions: models tuned to ANL tend to predict lower rates than those tuned to BNL. The MINERvA experiment [22] subsequently resolved much of this tension by simultaneously constraining the flux and the cross section.

K2K near detector (2008)

The K2K experiment [18] made the first accelerator-based total $\text{CC}\pi^+$ cross-section measurement on a carbon target at $\langle E_\nu \rangle = 1.3$ GeV using its fine-grained near detector. The result was consistent with NEUT predictions and established the nuclear-target correction from deuterium to carbon at intermediate energies.

MiniBooNE (ν_μ on CH_2 , BNB beam, 2011)

The MiniBooNE detector (818-tonne mineral-oil Cherenkov detector) published differential $\text{CC}\pi^+$ cross sections on CH_2 in the Booster Neutrino Beam ($E_\nu^{\text{peak}} \approx 0.7$ GeV) [19]. MiniBooNE was the first experiment to provide high-statistics differential distributions as a function of Q^2 , the invariant hadronic mass W , and the outgoing muon and pion kinematics. The flux-integrated total cross section was measured to be $\sigma \approx 3.4 \times 10^{-38}$ cm² per CH_2 molecule, or equivalently

$\sim 2.0 \times 10^{-38}$ cm² per carbon nucleus at $\langle E_\nu \rangle \approx 1$ GeV [26]. The measured rate exceeded the NUANCE prediction by $\sim 20\%$, i.e. that model *under*-predicted the data; the “generator over-prediction” discussed in the later literature refers to the opposite tension, in which GENIE-based predictions lie above subsequent measurements in parts of the phase space, and is generator- and phase-space-dependent rather than universal. The hydrogen content of CH₂ complicates the extraction of the carbon-only cross section, and the spherical Cherenkov geometry makes it difficult to measure low-energy pions or reconstruct proton recoils. These limitations drove subsequent adoption of finer-grained tracking detectors and liquid-argon targets.

MINERvA (ν_μ on CH, NuMI medium energy, 2015–)

The MINERvA experiment [20] uses a finely-segmented scintillator-strip tracking calorimeter in the NuMI medium-energy beam ($\langle E_\nu \rangle \approx 3.6$ GeV). It has published the most kinematically complete CC π^+ measurements to date, reporting differential cross sections as a function of: pion momentum p_π , pion angle θ_π , muon momentum p_μ , muon angle θ_μ , and the full set of Adler angles describing the pion in the hadronic centre-of-mass frame. MINERvA also measured CC π^+ on iron and lead targets [21], providing direct constraints on nuclear A -dependence at the same beam energy. GENIE consistently over-predicts MINERvA by 20–35% across all kinematic variables in the $\Delta(1232)$ peak, the same qualitative tension seen at lower energies. The higher beam energy at MINERvA means that the $p_\pi > 0.5$ GeV/ c region is well-populated, making it complementary to the MicroBooNE NuMI result which is statistics-limited at high pion momentum.

T2K ND280 (ν_μ on CH, off-axis beam, 2020)

T2K measured CC π^+ at its near detector ND280 in the off-axis T2K beam (OAB), which peaks at $\langle E_\nu \rangle \approx 0.85$ GeV [23]. The T2K OAB spectrum closely overlaps with the MicroBooNE NuMI off-axis spectrum in the kinematically-sensitive 0.3–1.5 GeV range, making this the most directly comparable prior measurement to the present work. T2K reported differential cross sections in p_μ and $\cos \theta_\mu$, finding a data/MC deficit of $\sim 30\%$ relative to NEUT. The carbon target ($A = 12$) differs from argon ($A = 40$), but the overlapping beam spectrum enables direct comparison of nuclear-mass dependence at fixed neutrino energy.

ArgoNEUT ($\nu_\mu + \bar{\nu}_\mu$ on Ar, NuMI, 2018)

ArgoNEUT [24] was a 175-litre LArTPC placed upstream of the MINOS near detector in the NuMI low-energy beam ($\langle E_\nu \rangle \approx 9.6$ GeV). Using 1.25×10^{20} POT, ArgoNEUT selected 115 ν_μ and 158 $\bar{\nu}_\mu$ CC π^+ candidate events after background subtraction [24], publishing the first CC π^+ total cross section on argon as a function of neutrino energy. The result was consistent with GENIE predictions at these high energies, where sensitivity to the $\Delta(1232)$ resonance is suppressed. The LArTPC technology and argon target make ArgoNEUT the direct predecessor to MicroBooNE, but the factor ~ 10 higher beam energy means that extrapolation to the $E_\nu < 2$ GeV regime relevant for oscillation experiments requires generator assumptions and limits the constraining power on FSI models in the resonance region.

MicroBooNE BNB channel: inclusive precedent (2022)

The same MicroBooNE detector used in this analysis published the first measurement of energy-dependent inclusive ν_μ charged-current cross sections on argon in the Booster Neutrino Beam [9] ($E_\nu^{\text{peak}} \approx 0.7$ GeV), establishing the detector’s absolute cross-section capability on argon together with the BNB flux modelling and reconstruction performance relevant to the present work. The dedicated exclusive CC π^+ measurement in the same beam followed on the full BNB dataset

(below), and it is that exclusive result to which the present NuMI CC π^+ measurement is most directly compared.

MicroBooNE BNB channel: full dataset (2026)

Reference [26] reports CC π^+ differential cross sections using the complete MicroBooNE BNB dataset of 1.11×10^{21} POT [26], selecting 6816 events after background subtraction and measuring differential cross sections in muon momentum, muon angle, pion momentum, and pion angle – the same four observables as the present analysis. The flux-integrated total cross section is

$$\sigma = (3.75 \pm 0.07_{\text{stat}} \pm 0.80_{\text{syst}}) \times 10^{-38} \text{ cm}^2/\text{Ar}$$

at mean neutrino energy $\langle E_\nu \rangle \approx 0.8$ GeV. Most generators reproduce the data within uncertainties; a notable exception is the muon angle with respect to the beam, where all tested generators overestimate the cross section at very forward angles ($\cos \theta_\mu \rightarrow 1$) [26]. The pion momentum differential cross section reported in this analysis represents the first such measurement on an argon target.

The BNB and NuMI analyses probe complementary phase space: BNB is on-axis with a narrow ν_μ spectrum peaked at ~ 700 MeV and negligible $\bar{\nu}_\mu$ content, while NuMI FHC at 135 mrad off-axis provides a softer, broader spectrum with a $\sim 35\%$ $\bar{\nu}_\mu$ admixture and extends sensitivity to $E_\nu \sim 0.1\text{--}0.5$ GeV.

MicroBooNE NuMI $\nu_e + \bar{\nu}_e$ CC π^+ (2025)

Reference [25] presents the first measurement of $\nu_e + \bar{\nu}_e$ CC single charged-pion differential cross sections on argon, using the full MicroBooNE NuMI dataset (2.0×10^{21} POT, FHC and RHC combined). The observable set – electron energy E_e , electron angle θ_e , pion angle θ_π , and electron–pion opening angle $\theta_{e\pi}$ – mirrors the muonic observables of the present analysis. The flux-integrated total cross section is measured to be

$$\sigma = (0.93 \pm 0.13_{\text{stat}} \pm 0.27_{\text{syst}}) \times 10^{-39} \text{ cm}^2/\text{nucleon}$$

at a mean ν_e energy of 730 MeV, in good agreement with all tested generator predictions (GENIE, NuWro, NEUT, GiBUU) within the combined uncertainties. Prior to this measurement, CC π^+ production by electron neutrinos on argon had never been measured; the result is a key input to the DUNE ν_e -appearance analysis, where CC π^+ events are a background to the ν_e quasi-elastic signal.

References

- [1] C. Andreopoulos *et al.*, “The GENIE Neutrino Monte Carlo Generator,” *Nucl. Instrum. Methods A* **614**, 87–104 (2010).
- [2] MicroBooNE Collaboration, P. Abratenko *et al.*, “New CC0 π GENIE model tune for MicroBooNE,” *Phys. Rev. D* **105**, 072001 (2022), arXiv:2110.14028 [hep-ex].
- [3] L. Alvarez-Ruso *et al.* (GENIE Collaboration), “Recent highlights from GENIE v3,” *Eur. Phys. J. ST* **230**, 4449–4467 (2021).
- [4] T. Golan, J. T. Sobczyk, and J. Żmuda, “NuWro: the Wrocław Monte Carlo generator of neutrino interactions,” *Nucl. Phys. B Proc. Suppl.* **229–232**, 499 (2012).
- [5] O. Buss *et al.*, “Transport-theoretical description of nuclear reactions,” *Phys. Rept.* **512**, 1–124 (2012); U. Mosel, “Neutrino event generators: foundation, status and future,” *J. Phys. G* **46**, 113001 (2019).

- [6] Y. Hayato and L. Pickering, “The NEUT neutrino interaction simulation program library,” *Eur. Phys. J. ST* **230**, 4469–4481 (2021).
- [7] D. Rein and L. M. Sehgal, “Coherent π^0 production in neutrino reactions,” *Nucl. Phys. B* **223**, 29–44 (1983).
- [8] C. Berger and L. M. Sehgal, “Partially conserved axial vector current and coherent pion production by low energy neutrinos,” *Phys. Rev. D* **79**, 053003 (2009).
- [9] MicroBooNE Collaboration, P. Abratenko *et al.*, “First Measurement of Energy-Dependent Inclusive Muon-Neutrino Charged-Current Cross Sections on Argon with the MicroBooNE Detector,” *Phys. Rev. Lett.* **128**, 151801 (2022), arXiv:2110.14023 [hep-ex].
- [10] D. Rein and L. M. Sehgal, “Neutrino Excitation of Baryon Resonances and Single Pion Production,” *Ann. Phys.* **133**, 79–153 (1981).
- [11] K. S. Kuzmin, V. V. Lyubushkin, and V. A. Naumov, “Lepton polarization in neutrino nucleon interactions,” *Mod. Phys. Lett. A* **19**, 2815–2829 (2004).
- [12] K. M. Graczyk and J. T. Sobczyk, “Form factors in the quark resonance model,” *Phys. Rev. D* **77**, 053001 (2008) [Erratum: *Phys. Rev. D* **79**, 079903 (2009)].
- [13] C. Berger and L. Sehgal, “Lepton mass effects in single pion production by neutrinos,” *Phys. Rev. D* **76**, 113004 (2007).
- [14] J. A. Nowak (MiniBooNE Collaboration), “Four-momentum transfer discrepancy in the charged current π^+ production in the MiniBooNE: Data vs. theory,” *AIP Conf. Proc.* **1189**, 243–248 (2009).
- [15] Y. Hayato, “A neutrino interaction simulation program library NEUT,” *Acta Phys. Polon. B* **40**, 2477–2489 (2009).
- [16] G. M. Radecky *et al.* (ANL Collaboration), “Study of single pion production by weak charged currents in low-energy νd interactions,” *Phys. Rev. D* **25**, 1161–1173 (1982).
- [17] T. Kitagaki *et al.* (BNL Collaboration), “Charged-current exclusive pion production in neutrino–deuterium interactions,” *Phys. Rev. D* **34**, 2554–2565 (1986).
- [18] A. Rodriguez *et al.* (K2K Collaboration), “Measurement of single charged pion production in the charged-current interactions of neutrinos in a 1.3 GeV wide band beam,” *Phys. Rev. D* **78**, 032003 (2008).
- [19] A. A. Aguilar-Arevalo *et al.* (MiniBooNE Collaboration), “Measurement of ν_μ -induced charged-current single positive pion production cross sections on mineral oil at $E_\nu \in 0.5$ –2.0 GeV,” *Phys. Rev. D* **83**, 052007 (2011).
- [20] B. Eberly *et al.* (MINERvA Collaboration), “Charged pion production in ν_μ interactions on hydrocarbon at $\langle E_\nu \rangle = 4.0$ GeV,” *Phys. Rev. D* **92**, 092008 (2015).
- [21] C. L. McGivern *et al.* (MINERvA Collaboration), “Cross sections for ν_μ and $\bar{\nu}_\mu$ induced pion production on hydrocarbon in the few-GeV region using MINERvA,” *Phys. Rev. D* **94**, 052005 (2016).
- [22] MINERvA Collaboration, J. Wolcott *et al.*, “Evidence for Quasielastic Charge Current Antineutrino Scattering Off Protons in Hydrocarbon at $\langle E_{\bar{\nu}} \rangle = 3.5$ GeV,” *Phys. Rev. Lett.* **116**, 081802 (2016).

- [23] K. Abe *et al.* (T2K Collaboration), “Measurement of the muon neutrino charged-current single π^+ production on hydrocarbon using the T2K off-axis near detector ND280,” *Phys. Rev. D* **101**, 012007 (2020), arXiv:1909.03936 [hep-ex].
- [24] R. Acciarri *et al.* (ArgoNEUT Collaboration), “First measurement of the cross section for ν_μ and $\bar{\nu}_\mu$ induced single charged pion production on argon using ArgoNEUT,” *Phys. Rev. D* **98**, 052002 (2018), arXiv:1804.10294 [hep-ex].
- [25] MicroBooNE Collaboration, P. Abratenko *et al.*, “First measurement of ν_e and $\bar{\nu}_e$ charged-current single charged-pion production differential cross sections on argon using the MicroBooNE detector,” *Phys. Rev. Lett.* **135**, 061802 (2025), arXiv:2503.23384 [hep-ex].
- [26] MicroBooNE Collaboration, P. Abratenko *et al.*, “Measurement of single charged pion production in charged-current ν_μ -Ar interactions with the MicroBooNE detector,” *Phys. Rev. D* **113**, 032007 (2026), arXiv:2509.03628 [hep-ex].